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Time-stratigraphic framework for the Eocene-Oligocene Mogollon-Datil volcanic field, southwest New Mexico

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ABSTRACT

A time-stratigraphic framework for discontinuously exposed regional ignimbrites in the Eocene-Oligocene Mogollon-Datil volcanic field has been established using correlations aided by $^{40}Ar/^{39}Ar$ age determinations and paleomagnetic analyses. $^{40}Ar/^{39}Ar$ age spectra from sanidine separates (25 regional ignimbrites, 85 samples, 97 spectra) yield well-defined plateau ages that are precise (within-sample and within-unit $1\sigma < \pm 0.5\%$) and agree closely with independently established stratigraphic order. Paleomagnetic remanence directions (404 sites) from individual ignimbrite outflow sheets are generally vertically and horizontally uniform throughout facies ranging from thick (100–500 m), densely welded, proximal ignimbrites to thin (1.5–30 m), unwelded, distal fringes. Between-unit differences in paleomagnetic directions provide useful correlation criteria, particularly for units having ages too close to be resolved using $^{40}Ar/^{39}Ar$ dating.

The Mogollon-Datil time-stratigraphic framework clarifies ignimbrite history and provides improved age control for intercalated lavas and sedimentary rocks. Ignimbrite activity was strongly episodic; outflow sheets were primarily erupted in four discrete pulses representing synchronized activity of two separate cauldron complexes. Activity in the southern complex began at 36.2 Ma near Las Cruces, New Mexico, and subsequently migrated 220 km northwest, culminating in the 28.0 Ma Bursum cauldron. Activity in the northern complex, located west of Socorro, New Mexico, underwent a less defined and more modest 40-km westward migration over its 32.0 to 24.3 Ma life span. The four pulses of ignimbrite activity were (1) 36.2–24.3 Ma, 12 major units, >1,500 km^3 total volume; (2) 32.0–31.4 Ma, three major units, >1,500 km^3 volume; (3) 29.1–27.4 Ma, nine major units, >6,000 km^3; and (4) 24.3 Ma, one major unit. The third and largest ignimbrite pulse was accompanied by extensive rhyolitic dome and flow eruptions in the area between the two main cauldron complexes.

INTRODUCTION

Since the late 1950s, the 40,000-km^2 late Eocene-Oligocene Mogollon-Datil volcanic field (Figs. 1 and 2) has been the focus of numerous mapping, stratigraphic, geochemical, petrologic, and isotopic studies, primarily by workers from New Mexico universities, the New Mexico Bureau of Mines and Mineral Resources, and the U.S. Geological Survey. Detailed geologic mapping has been successfully used to unravel the stratigraphy in parts of the field (for example, Ratté and others, 1984; Osburn

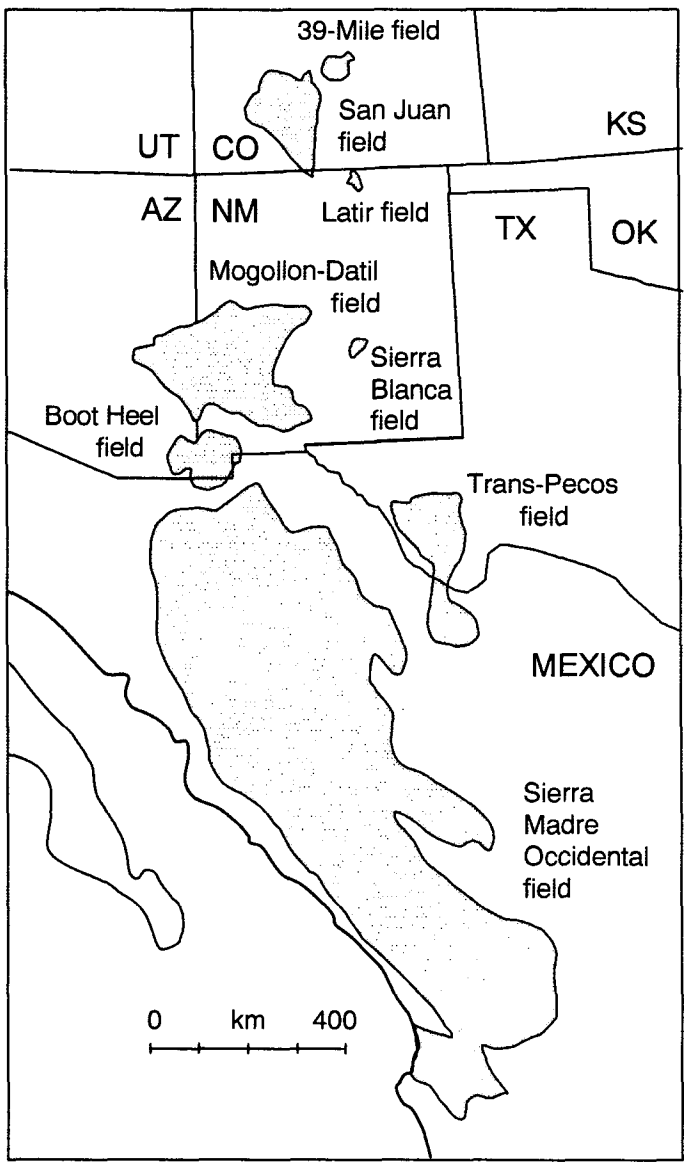


Figure 1. Generalized map of the present-day extent of mid-Tertiary silicic volcanic fields of southwestern North America.

Geological Society of America Bulletin, v. 104, p. 851–871, 12 figs., 1 table, July 1992.

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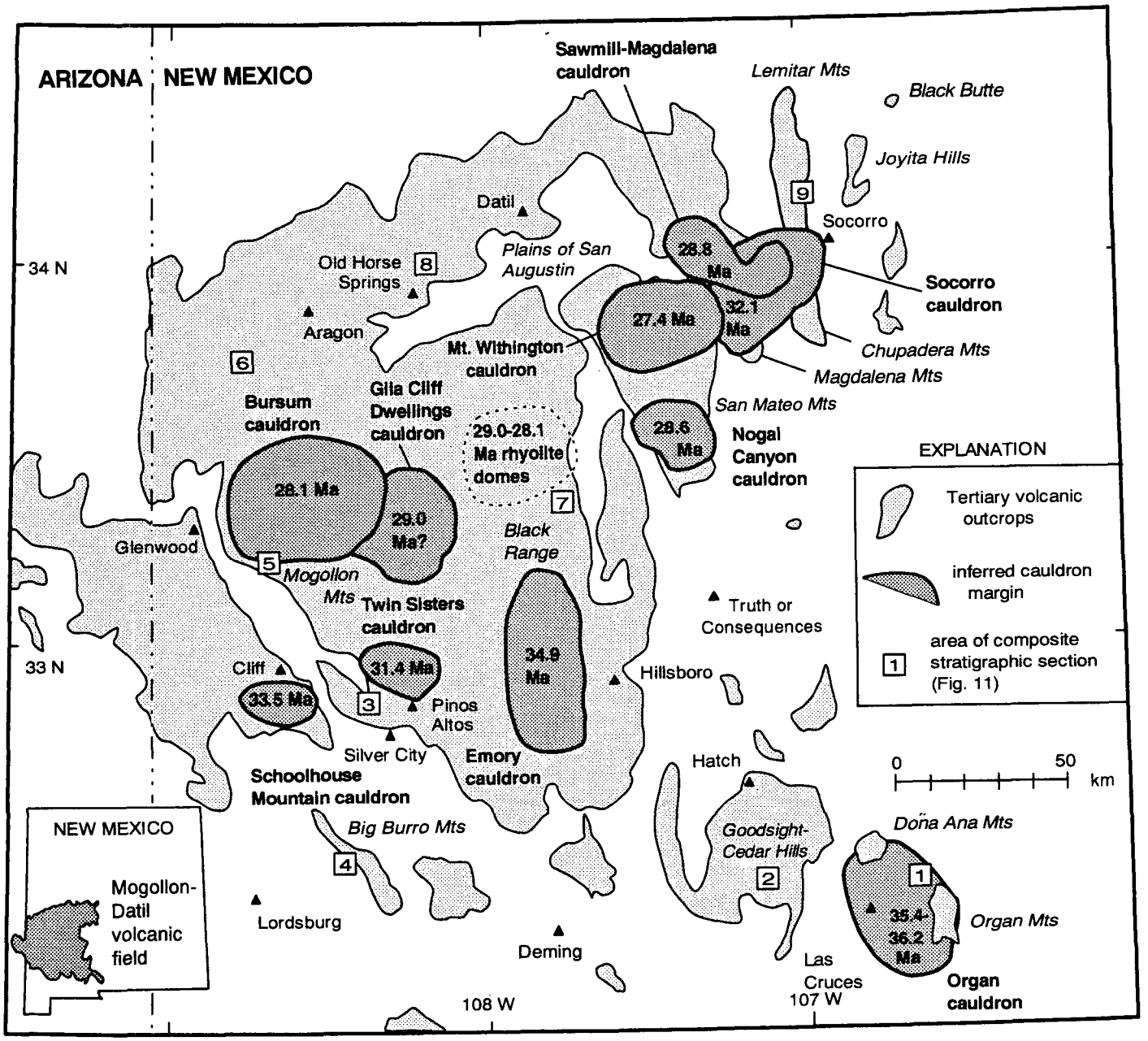


Figure 2. Map of the Mogollon-Datil volcanic field depicting inferred cauldrons, their ages, and generalized outcrop distribution of late Eocene-Oligocene volcanic rocks. Numbered squares show general locations of composite stratigraphic sequences presented in Figure 11.

and Chapin, 1983a, 1983b; Seager and others, 1982), and K-Ar studies have helped to resolve the ages of many units (Elston and others, 1973; Osburn and Chapin, 1983b; Marvin and others, 1987).

One persistent problem in understanding the geology of the volcanic field has been lack of an accurate, integrated, regional stratigraphic framework. The principal obstacle to this has been the inability to reliably correlate the abundant regional ignimbrites. This paper summarizes results of a study in which $^{40}Ar/^{39}Ar$ dating and paleomagnetism have been combined to address this problem. This report concentrates on the stratigraphy and eruptive history of the ignimbrites; no lithologic or geochemical data are reported, and no attempt is made to address the history of voluminous mafic to intermediate lavas associated with the ignimbrites. Other papers present in greater detail the $^{40}Ar/^{39}Ar$ dating work (McIntosh and others, 1990), paleomagnetic aspects of the study (McIntosh, 1991), and an unabridged $^{40}Ar/^{39}Ar$ and paleomagnetic data set (McIntosh and others, 1991).

GEOLOGIC SETTING

The late Eocene-Oligocene Mogollon-Datil volcanic field of southwestern New Mexico is part of a discontinuous belt of mid-Tertiary silicic volcanic fields (Fig. 1) that extends from the San Juan Mountains in

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Colorado southward into central Mexico. The boundaries of the Mogollon-Datil field are not formally defined. To the north and east, it is generally eroded; to the west, it is buried by younger volcanic rocks; and to the south, it merges with the "boot heel" area in the southwestern corner of New Mexico, where additional mid-Tertiary volcanic units are present in a series of isolated fault-block ranges (Fig. 1). For the purposes of this report, the Mogollon-Datil field is considered to extend as far south as Las Cruces and Lordsburg, New Mexico (Fig. 2), thereby excluding New Mexico's "boot heel."

Mogollon-Datil activity followed a pattern common to several mid-Tertiary volcanic fields of southwestern north America; that is, initial intermediate volcanism was followed by cauldron-forming silicic activity (Lipman and others, 1972, 1978). In the Mogollon-Datil field, basaltic andesite to andesite volcanoes were active from about 40 to 36 Ma, followed by episodic bimodal basaltic andesite and silicic activity from 36 to 24 Ma (Elston, 1984; Cather and others, 1987; McIntosh and others, 1990). Most proposed models explain the mid-Tertiary volcanism of the southwestern United States as a consequence of a transient episode of low-angle subduction of the Farallon plate (for example, Lipman and others, 1972; Cross and Pilger, 1982). Alternative models interpret magmatism as a result of uplifted isotherms following post-Laramide crustal thinning (Mutschler and others, 1987).

From mid-Oligocene time to the present, the Mogollon-Datil volcanic field has experienced inhomogeneously distributed extensional tectonism (Chapin and Seager, 1975). As a result, originally contiguous ignimbrite sheets are now discontinuously exposed in fault-block mountain ranges separated by infilled basins. Structural complexity varies widely within the area, ranging from unfaulted, subhorizontal strata to zones of low-angle faulting and steep (45°–90°) dips indicative of extension in excess of 100% (Chamberlin, 1983).

Silicic units in the Mogollon-Datil field include domes, flows, intrusions, and numerous ignimbrites. Individual ignimbrites vary widely in form, volume, and distribution, ranging from cauldron-derived, densely welded outflow sheets as large as 1,250 km^3 to tiny (<0.1-km^3), unwelded pyroclastic aprons surrounding rhyolite domes. Units sampled in this study (Table 1) include 25 regional ignimbrites (arbitrarily defined as >40-km known horizontal extent) and 54 smaller units.

Regional ignimbrites constitute potentially ideal time-stratigraphic marker horizons for the Mogollon-Datil volcanic field because they were emplaced almost instantaneously. Prior to this study, ignimbrite stratigraphy in several areas of the volcanic field had already been established, using detailed mapping and correlations based on lithology and stratigraphic position relative to other units (for example, Ratté and others, 1984; Osburn and Chapin, 1983b; Seager and others, 1982). Regional correlations, however, had been problematic. Correlations based solely on mapping are hampered by discontinuous outcrop patterns produced by basin-and-range extension. Lithology, although useful as a first-order criterion, is insufficient for long-distance correlations. Many stratigraphically distinct units are lithologically similar, and individual units have lateral, vertical, and sectoral variations in mineralogy, chemistry, welding texture, and phenocryst concentration and size, analogous to variations observed in ignimbrites elsewhere (Hildreth and Mahood, 1985). Specific ignimbrites become difficult to identify outside of their familiar stratigraphic context, particularly at their thin, unwelded distal fringes, where they may be easily confused with local rhyolite-dome–derived units. Attempts to fingerprint ignimbrites geochemically have been largely unsuccessful (Bornhorst, 1980), and conventional K-Ar dating techniques lack the resolution required by most of the correlation problems (for example, Marvin and others, 1987). Published K-Ar dates from Mogollon-Datil ignimbrites generally range from 36 to 24 Ma, but multiple dates obtained from individual units typically vary by ±10% or more (Elston and others, 1973; Chapin and others, 1975; Marvin and others, 1987), and some of the K-Ar data contradict established stratigraphic order (for example, Clemons, 1976).

In addition to difficulties in correlating Mogollon-Datil outflow sheets, there have been problems identifying their source cauldrons. Source cauldrons have been reliably determined for only seven of the major ignimbrites, and other cauldrons have been identified for which outflow sheets are unknown (Table 1 and Fig. 2; Osburn and Chapin, 1983a; Elston, 1984; Seager and McCurry, 1988; Ratté and others, 1984; Wahl, 1980).

METHODS

Mogollon-Datil ignimbrites are well suited to paleomagnetic and $^{40}Ar/^{39}Ar$ analysis. High relief and the arid New Mexico climate provide well-exposed, unweathered outcrops in which alteration is generally minimal, although K-metasomatism or hydrothermal alteration has affected some areas (Chapin and Lindley, 1986). Most of the units contain abundant sanidine, a K-rich, Ar-retentive, anhydrous mineral well suited to high-precision $^{40}Ar/^{39}Ar$ dating (Zeitler, 1987; McIntosh and others, 1990). Finally, existing detailed mapping and stratigraphic studies offer a context in which these techniques can be tested and applied.

$^{40}Ar/^{39}Ar$ Dating

The $^{40}Ar/^{39}Ar$ method is becoming increasingly popular for dating mid-Tertiary volcanic rocks, chiefly because it can yield reliable age determinations which are several times more precise than conventional K-Ar or fission-track data (for example, Lanphere, 1988; Dalrymple and Duffield, 1988). Advantages include the capability to simultaneously measure radiogenic ^{40}Ar and ^{40}K (as reactor-produced ^{39}Ar) on a single aliquot of sample material (Turner, 1968; Roddick, 1983), and the capability to accurately measure ages without extracting all Ar from a sample, thereby overcoming the chronic problem of incomplete ^{40}Ar extraction in K-Ar dating of sanidine and anorthoclase (McDowell, 1983).

Presently, the two most precise methods of $^{40}Ar/^{39}Ar$ dating involve laser fusion of single mineral grains (for example, Dalrymple and Duffield, 1988; Deino, 1989) and, as in this study, obtaining age spectra by incremental heating of multigrain mineral separates. Age spectra can help to identify effects of ^{40}Ar loss, excess ^{40}Ar, or contamination by older material, although the single-grain, laser-fusion method is more sensitive to contaminant grains (Lippolt and others, 1986; Bogaard and others, 1987; Lo Bello and others, 1987).

$^{40}Ar/^{39}Ar$ dating procedures and results are detailed in McIntosh and others (1990, 1991). In brief, a total of 85 samples from 36 ignimbrites were analyzed by this method. Sanidine separates were prepared and irradiated, along with flux monitors of known age (FCT–3, 27.83 Ma, Kunk and others, 1985; Mmhb1, 520.4 Ma, Samson and Alexander, 1983), in the U.S. Geological Survey TRIGA reactor (Dalrymple and others, 1981). Age spectra were measured at the U.S. Geological Survey $^{40}Ar/^{39}Ar$ dating facility at Reston, Virginia. Spectra are characteristically flat (Fig. 3); plateau ages were calculated by the method of Fleck and others (1977). Replicate plateau age determinations (n = 2 to 6) for eight different samples have within-sample 1σ values averaging ± 0.25%. Plateau ages from multiple (n = 3 to 8) samples of individual ignimbrites have within-unit 1σ values of ± 0.1–0.4% (± 0.04–0.13 Ma). In sharp contrast, conventional K-Ar and fission-track ages on the same ignimbrites gener-

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TABLE 1. SUMMARY OF $^{40}Ar/^{39}Ar$ AND PALEOMAGNETIC DATA FOR MOGOLLON-DATIL IGNIMBRITES

Stratigraphic unit	Abbreviation	Reference	Extent	Volume	$^{40}Ar/^{39}Ar$ age			Paleomagnetic remanence					Cauldron or source
			(km)	(km^3)	n	Ma	σ	polarity	n/nt	inc	dec	α_{95}	
Episode 4													
tuff of Turkey Springs	TS	F86	62		2	24.33	±0.03	R	15/19	-51.8	175.9	3.6	
tuff of Slash Ranch	SR	R78	22		1	26.10	±0.12	N	2/3	57.6	166.9	13.0	
Episode 3													
South Canyon Tuff	SC	OC83	135	700	3	27.36	±0.07	R	22/43	-30.7	202.1	3.5	Mt. Withington
tuff of Walking X Canyon	WX	H78			1	27.58	±0.10	N	1/1	64.0	5.7		
Lemitar Tuff	LM	OC83	78	450	2	28.00	±0.08	N	10/21	36.6	356.9	4.3	
tuff of Caronita Canyon	CC	OC83	40	50	1	27.96	±0.07	N	4/5	41.5	354.2	11.0	
tuff of Triangle C Ranch	TC	R89	58		1	28.05	±0.14	N	5/6	49.2	357.3	4.4	Bursum?
Apache Springs Tuff	AS	R84	35	1200	1	27.98	±0.09	R	2/2	-35.8	166.1	30.6	Bursum
Bloodgood Canyon Tuff	BG	R84	155	1000	8	28.05	±0.04	R	22/23	-27.1	165.3	3.1	Bursum
tuff of Diablo Range	DR	RG75	10		1	28.05	±0.08	R	1/1	-24.8	200.3		dome
Shelley Peak Tuff	SP	R84	96	100-200		(28.1)		R	8/10	-32.7	174.8	6.4	Gila Cliff Dwellings?
tuff of Garcia Camp	GC	D87	25		1	28.10	±0.09	R	7/7	-48.8	185.4	5.2	dome
Vicks Peak Tuff	VP	OC83	200	1050	3	28.56	±0.04	R	24/31	-27.9	182.8	3.3	Nogal Canyon
tuff of Lookout Mountain	LO	W82			2	28.69	±0.03	R	1/1	-55.4	160.8		dome
La Jencia Tuff	LJ	OC83	158	1250	4	28.85	±0.04	R	15/24	-49.3	168.2	3.6	Sawmill/Magdalena
Davis Canyon Tuff	DC	R84	125	200-400	2	29.01	±0.11	R	6/7	-53.5	159.6	7.5	Gila Cliff Dwellings?
tuff of Mud Hole	MH	H90			1	29.09	±0.11	R	1/1	-55.9	124.5		dome?
tuff of Stiver Canyon		W82				(29)		R	1/1	-52.7	172.6		dome?
tuff of Little Mineral Creek	LC	H90			1	29.01	±0.10	N	1/1	58.0	3.1		dome
tuff of Monument Canyon		K81	25			(29)		N	1/1	35.2	344.0		
Episode 2													
Tadpole Ridge Tuff, upper member	TP	F76	28		1	31.39	±0.11	R	3/3	-23.3	185.2	16.3	Twin Sisters
Tadpole Ridge Tuff, lower member		F76	45			(31.4)		R	6/6	-71.1	203.8	11.0	Twin Sisters
Caballo Blanco Tuff	CB	S82	143		5	31.65	±0.06	R	7/9	-46.0	171.4	4.4	
Hells Mesa Tuff	HM	OC83	152	1200	4	32.06	±0.10	R	10/18	-60.2	166.0	4.8	Socorro
Episode 1													
Box Canyon Tuff	BX	E57	225	>100	7	33.51	±0.13	R	12/17	-56.0	141.3	4.7	
Blue Canyon Tuff	BC	OC83	135		2	33.66	±0.03	R	6/9	-37.3	144.4	5.7	
tuff of Table Mountain		E57			1	33.89	±0.12	R	1/1	-36.9	181.5		
Cooney Tuff	CN	R84	70?			(34)		R	3/3	-46.7	168.7	28.0	
Rock House Canyon Tuff	RH	OC83	142		1	34.42	±0.12	R	8/11	-55.6	144.8	5.4	
Mimbres Peak Tuff	?	E75			1	34.57	±0.12	N	1/1	58.7	353.0		dome
tuff of Lebya Well	LW	RE89	72		1	34.70	±0.10	R	4/4	-61.7	171.2	10.8	
Kneeling Nun Tuff	KN	E75	213	>900	4	34.89	±0.05	R	21/28	-43.7	167.1	3.8	Emory
tuff of Stone House Ranch		H90				(34.9)		N	3/3	43.5	323.9	18.3	
Bell Top Formation, tuff member 4	B4	C76	112		3	34.96	±0.04	N	5/5	56.1	2.7	3.6	
Sugarlump Tuff	SL	S82	50?		1	35.17	±0.12	N	2/2	62.3	337.7	8.1	
tuff of Farr Ranch	FR	RE89	143		1	35.57	±0.13	N	9/13	50.4	345.5	4.7	
Datil Well Tuff	DW	OC83	140		2	35.48	±0.07	N	5/5	53.3	19.9	6.4	
Bell Top Formation, tuff member 3	B3	C76	41		1	35.69	±0.12	N	2/2	69.9	329.9	1.6	Organ?
Bell Top Formation, tuff member 2	B2	C76	15			(36)		N	2/2	39.7	350.0	17.3	
Doña Ana Tuff	DA	S76	8		1	35.49	±0.11	N	2/2	54.4	155.3	48.5	Doña Ana/Organ
Squaw Mountain Tuff	SM	SM88	11		1	35.75	±0.12	N	2/2	58.8	178.0	44.5	Organ
Achenback Park Tuff	AP	SM88	13			(35.8)		N	3/3	50.0	177.0	35.6	Organ
Cueva Tuff	CT	SM88	13		1	36.20	±0.13	R	2/3	-53.4	220.3	67.8	Organ

Explanation: abbreviations refer to units shown in Figure 11; extent is maximum exposed lateral dimension (extents based on uncertain correlations are queried); volume estimates are after McIntosh and others, 1986; $^{40}Ar/^{39}Ar$ age is mean of plateau age determinations; n is number of analyses used in mean; σ is standard deviation of multiple ages or analytical error of single ages; paleomagnetic remanence shows unit-mean data; n/nt is number of sites used in unit-mean/total sites; inc, dec, and α_{95} are unit-mean inclination, declination, and cone of 95% confidence (α_{95} omitted for units with results from only one site); source cauldrons are shown where known, dome denotes small-volume units derived from rhyolitic domes.

References: C76 = Clemons, 1976; D87 = Duffield and others, 1987; E57 = Elston, 1957; E75 = Elston and others, 1975; F76 = Finnell, 1976; F86 = Ferguson, 1986; H78 = Hedlund, 1978; H90 = Harrison, 1990; K81 = Krier, 1981; OC83 = Osburn and Chapin, 1983b; R78 = Richter, 1978; R84 = Ratté and others, 1984; R89 = Ratté, 1989; RE89 = Ratté and others, 1989; RG75 = Ratté and Gaskill, 1975; S76 = Seager and others, 1976; S82 = Seager and others, 1982; SM88 = Seager and McCurry, 1988; W82 = Woodard, 1982.

ally have within-unit 1σ values in excess of ± 5% (± 1.3–2.2 Ma) (Marvin and others, 1987; Ratté and others, 1984). A further indication of the high precision of the $^{40}Ar/^{39}Ar$ age determinations is their close agreement with established stratigraphic order, as presented in McIntosh and others (1990) and discussed below.

Paleomagnetism

Paleomagnetism has been successfully used as an inexpensive, reliable, ignimbrite correlation criterion for more than two decades (for example, Dalrymple and others, 1965; Grommé and others, 1972; Reynolds, 1977). Many ignimbrites, both welded and non-welded, possess thermoremanent magnetization (TRM) directions representing an accurate record of the geomagnetic field direction during rapid post-emplacement cooling (Dalrymple and others, 1965; Grommé and others, 1972; Best and others, 1973; Reynolds, 1977; Hoblitt and others, 1985; Hildreth and Mahood, 1985; Weiss and others, 1989). Due to reversals and secular variation of the geomagnetic field, paleomagnetic analysis can potentially distinguish between volcanic units differing in age by as little as centuries (Grommé and others, 1972; Bogue and Coe, 1981; Holcomb and others, 1986). Paleomagnetic analysis offers the further potential of determining tectonic rotations (Hagstrum and Gans, 1989; Wells and Hillhouse, 1989; Rosenbaum and others, 1991) and of placing precisely dated units within the established magnetic polarity time scale (McIntosh, 1988, 1989).

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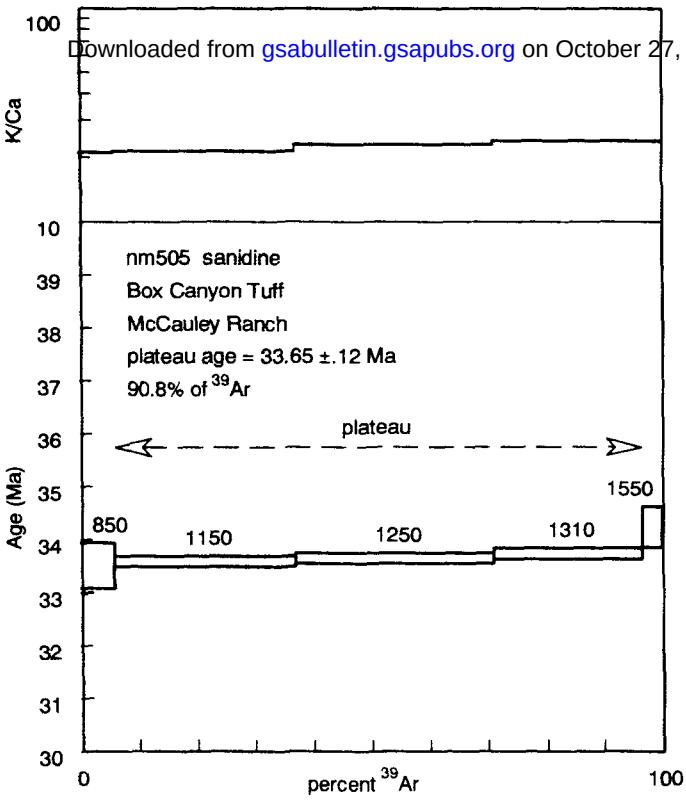
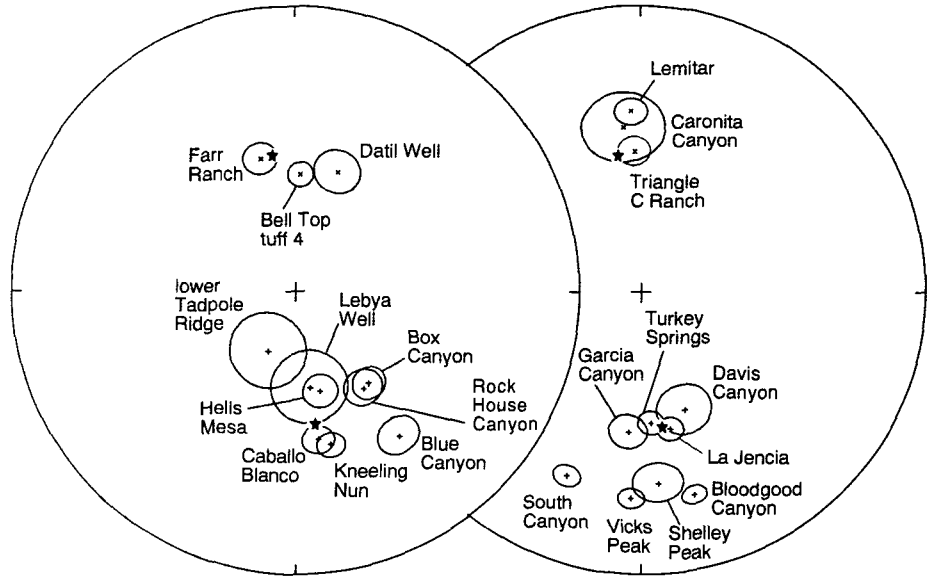


Figure 3. Representative $^{40}Ar/^{39}Ar$ age spectrum from sanidine separate of the Box Canyon Tuff. The nearly flat shape of this age spectrum is typical of results from more than 90% of the samples analyzed in this study.

Paleomagnetic procedures are detailed in McIntosh (1991) and McIntosh and others (1991). Altogether, 3,055 oriented samples were collected from 404 sites in 25 regional and 54 local ignimbrites; samples were primarily field drilled and oriented by suncompass. Remanent magnetizations (RMs) were determined by spinner and cryogenic magnetometers, utilizing standard alternating field and thermal demagnetization procedures. Magnetic mineralogy was assessed by light microscopy, thermomagnetic analyses, and isothermal remanent magnetization (IRM) acquisition and demagnetization experiments.

Mogollon-Datil ignimbrites generally carry uniform, well-defined RMs that provide reliable correlation criteria. About 90% of the sites have well-grouped ($\alpha_{95}<10°$) RMs carried by finely dispersed, high-temperature, deuteric-oxidation assemblages of magnetite, hematite, and maghemite. These RMs, interpreted as TRMs, are readily separated from abundant lightning-induced isothermal components. Furthermore, TRMs are present even in ignimbrites suffering moderate hydrothermal alteration or K-metasomatism, apparently reflecting their oxidation-resistant deuteric hematite (McIntosh, 1989, 1991). For the remaining 10% of sites, primarily in strongly altered or poorly welded, lithic-rich tuff, the original TRMs are obscured by chemical remanence or randomly directed, lithic-hosted RMs (McIntosh, 1989, 1991). Results from these sites are excluded from all following figures.

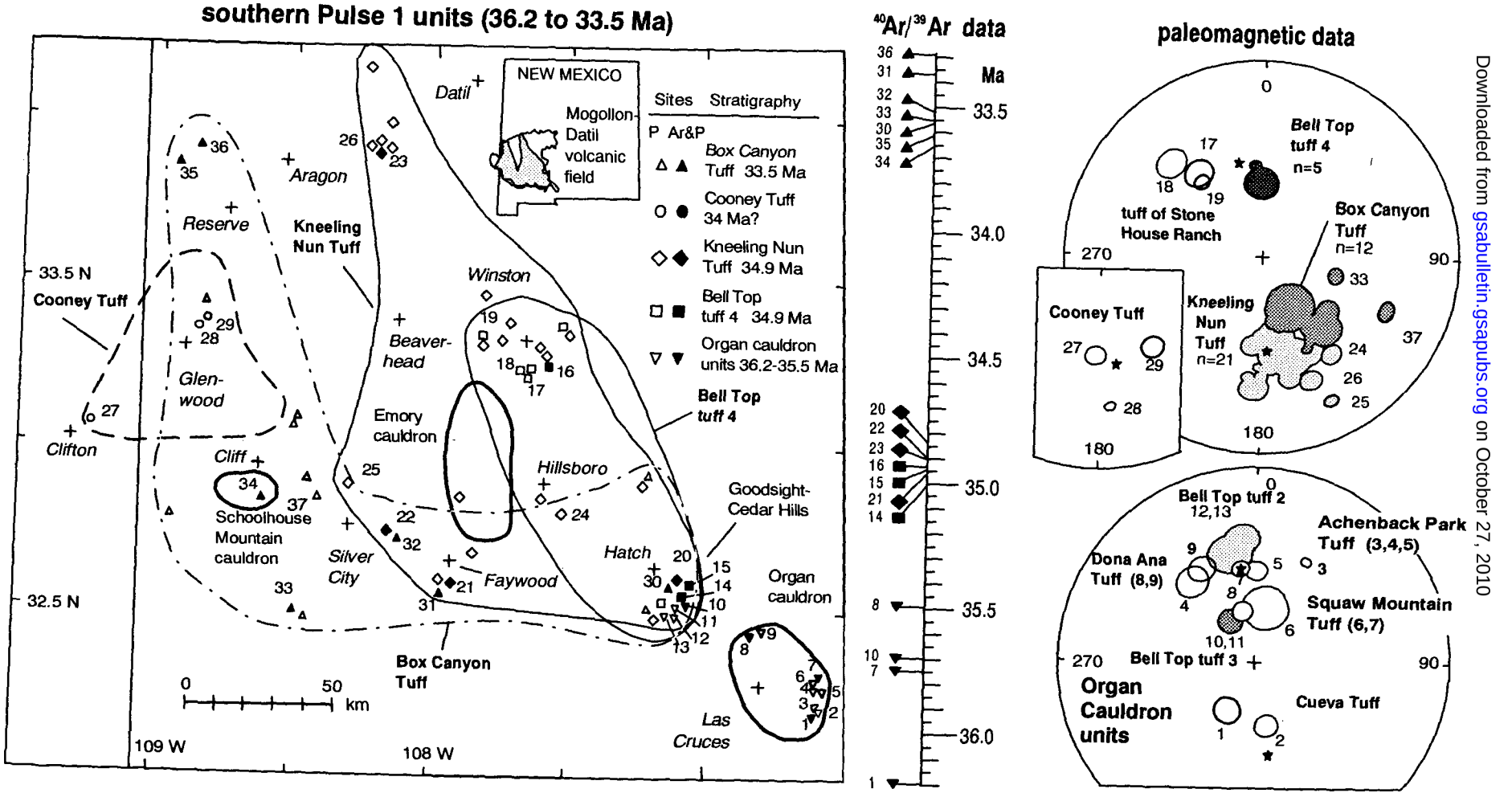
Within individual ignimbrite outflow sheets, TRM directions are generally laterally and vertically consistent, in both densely welded proximal and poorly welded distal facies. Some of the ignimbrite outflow sheets have one or more sites with discordant site-mean directions (arbitrarily defined as diverging from the unit-mean direction by >15°). These discordant sites primarily reflect local tectonic rotations and errors in determining the paleohorizontal (McIntosh, 1991). These sites were rejected from calculations of unit-mean directions (Fig. 4 and Table 1); they are individually identified in Figures 5–10 but are only discussed where pertinent to correlation problems. The magnetizations of the outflow



a) Well-constrained unit-mean directions 35.6-31.4 Ma

b) Well-constrained unit-mean directions 29.0-24.3 Ma

Figure 4. Unit-mean paleomagnetic directions for Mogollon-Datil ignimbrites. Each ellipse represents the cone of 95% confidence (α_{95}) for a unit-mean direction based on four or more sites. All Northern Hemisphere directions are downward and all Southern Hemisphere directions are upward. Stars represent time-averaged Eocene-Oligocene geomagnetic reference directions (Diehl and others, 1988). Table 1 presents unit-mean paleomagnetic data in tabular form. (a) 35.6 to 31.4 Ma units; (b) 29.0 to 24.3 Ma units.



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Figure 5. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 36.2 to 33.5 Ma pulse 1 ignimbrites in the southern Mogollon-Datil volcanic field.

Organ cauldron units. Paleomagnetic and $^{40}Ar/^{39}Ar$ data support interpretation of Bell Top tuff 3 as an outflow-facies equivalent of the co-eruptive intracaldera-facies Squaw Mountain and Achenback Park Tuffs.

Bell Top tuff 4. This unit extends into the Winston area and probably erupted from the Emory cauldron as a precursor to the Kneeling Nun Tuff. Paleomagnetic data indicate that the tuff of Stone House Ranch (Harrison, 1990) is not correlative.

Kneeling Nun Tuff. Correlations based on paleomagnetic and $^{40}Ar/^{39}Ar$ data indicate that the outflow sheet, erupted from the Emory cauldron, extends farther to the north and southeast than previously believed. Bell Top tuff 5 (Clemons, 1976) represents the southeastern distal fringe of the Kneeling Nun Tuff.

Cooney Tuff. This multi-cooling-unit ignimbrite was apparently erupted over a relatively long period, resulting in poorly grouped paleomagnetic directions. No $^{40}Ar/^{39}Ar$ ages have been obtained from this sanidine-poor unit.

Box Canyon Tuff. Correlations supported by paleomagnetic and $^{40}Ar/^{39}Ar$ data indicate a large regional extent for this outflow sheet, formerly interpreted as a local unit (Elston, 1957). Schoolhouse Mountain cauldron is the probable source of the ignimbrite.

In this figure, and in Figures 6 through 10, cauldrons are outlined by heavy dashed lines, and generalized areal extents of ignimbrite outflow sheets are shown by lighter lines. Open symbols denote paleomagnetic sites and filled symbols denote locations where both paleomagnetic and $^{40}Ar/^{39}Ar$ samples were collected. $^{40}Ar/^{39}Ar$ age determinations for individual samples are presented on the vertical time line; numbers correspond to numbered sites on the location map. Paleomagnetic data are displayed on equal-area stereographic projections. All northern hemisphere directions are downward and all southern hemisphere directions are upward. Concordant paleomagnetic data for each unit are depicted as irregular outlines which encompass individual site-mean α_{95} ellipses which fall within 15° of the unit-mean direction shown in Figure 4 and Table 1; locations for these sites are not numbered on the location map. Data for sites with discordant directions (>15° from their respective unit-mean direction) are shown as individual ellipses; adjacent numbers correspond to numbered sites on the location maps. Sites yielding imprecise (α_{95} > 10°) paleomagnetic data are excluded from stereographic projection and location map. Stars represent time-averaged Eocene-Oligocene geomagnetic reference directions (Diehl and others, 1988). Sites 16, 31, and 32 lack paleomagnetic data. Complete paleomagnetic and $^{40}Ar/^{39}Ar$ data are presented in McIntosh and others (1991).

sheets show no evidence of syncooling geomagnetic secular variation or sub-blocking-temperature flowage (unlike Rosenbaum, 1986).

Unit-mean TRM directions of all 22 well-studied (4 to 22 sites per unit) ignimbrite outflow sheets (Fig. 4) lie within 30° of the time-averaged late Eocene–Oligocene reference direction for southern New Mexico (D = 170.5, I = 50), calculated from paleomagnetic pole data of Diehl and others (1988). Nine of these 22 ignimbrites have distinctive unit-mean TRM directions which differ from the reference direction by more than 15°; paleomagnetic data from these units tend to provide stronger correlation criteria than data from paleomagnetically less distinctive units.

Compared to outflow sheet data, the site-mean TRMs of thick (>500 m) intracauldron facies ignimbrites are poorly grouped, apparently reflecting either secular variation during protracted cooling or large discrepancies between eutaxitic foliations and the syncooling paleohorizontal (McIntosh, 1991).

AGE, CORRELATION, AND DISTRIBUTION OF MOGOLLON-DATIL IGNIMBRITES

The combination of $^{40}Ar/^{39}Ar$ age determinations and paleomagnetic data has proven valuable for correlating Mogollon-Datil ignimbrites. In cases where intervals between eruptions of lithologically similar ignimbrites were relatively long (≥0.5 m.y.), units can generally be identified using $^{40}Ar/^{39}Ar$ data alone. In cases where eruptions were more frequent, differences in paleomagnetic polarity or TRM direction become more important for distinguishing units. In general, $^{40}Ar/^{39}Ar$ and paleomagnetic criteria have tended to be complementary. $^{40}Ar/^{39}Ar$ data have helped to resolve age differences between units that have similar site-mean paleomagnetic directions or paleomagnetic data disturbed by tectonic rotations. Likewise, paleomagnetic data have helped to distinguish units that lack fresh sanidine, or are too close in age to be resolved by the $^{40}Ar/^{39}Ar$ method.

Two fundamental features emerge from the ignimbrite data set. First, $^{40}Ar/^{39}Ar$ ages fall into four distinct groups representing strongly episodic pulses of ignimbrite volcanism (Table 1). Second, the regional ignimbrites were erupted from two distinct areas: a northern cluster of cauldrons south and west of Socorro, and a southern band of cauldrons extending from Las Cruces to Glenwood (Fig. 2). The following sections present age and distribution data for individual Mogollon-Datil ignimbrites, followed by a general discussion of Mogollon-Datil volcanism. Paleomagnetic and $^{40}Ar/^{39}Ar$ data are summarized in Table 1 and Figures 5 to 10; more complete data are presented in McIntosh and others (1991).

Pulse 1 (36.2 to 33.5 Ma): First Flareup

The first pulse of Mogollon-Datil ignimbrite volcanism spanned the interval between 36.2 and 33.5 Ma. Pulse 1 ignimbrites, primarily low-silica rhyolite in composition, include 11 regional outflow sheets and several smaller, apparently local units (Table 1 and Figs. 5 and 6). The regional units consist of two large-volume units (Kneeling Nun Tuff, 34.9 Ma, >900 km^3; Box Canyon Tuff, 33.5 Ma, >500 km^3) and several other thin (5–50 m) but widely dispersed (1,000–10,000 km^2) ignimbrites. The unusually low aspect ratio (Walker and others, 1980) of these thin ignimbrites probably reflects some combination of subdued pre-eruptive topography and high energy or high mobility of the ash flows.

The following sections consider pulse 1 ignimbrites in chronological order. Many of these units are poorly exposed; therefore, some of the proposed correlations are considered tentative. Only two cauldrons have been identified for pulse 1 ignimbrites.

Pre–Kneeling Nun Tuff Units (36.2 to 35.0 Ma). Both northern and southern eruptive centers were active prior to eruption of the Kneeling Nun Tuff at 34.9 Ma; identified pre–Kneeling Nun ignimbrites include seven distinct outflow sheets and one intracauldron sequence.

The oldest known Mogollon-Datil ignimbrites are intracauldron-facies units exposed in the southeastern corner of the volcanic field (Fig. 5). Detailed mapping in the Organ Mountains (Seager, 1981; Seager and others, 1976) delineated the eastern margin of a cauldron that contains a 9,000-m-thick ignimbrite sequence (Cueva, Achenback Park, and Squaw Mountain Tuffs, in ascending order) which has coherent normal geochemical zoning and K-Ar ages near 33 Ma (Seager and McCurry, 1988). The

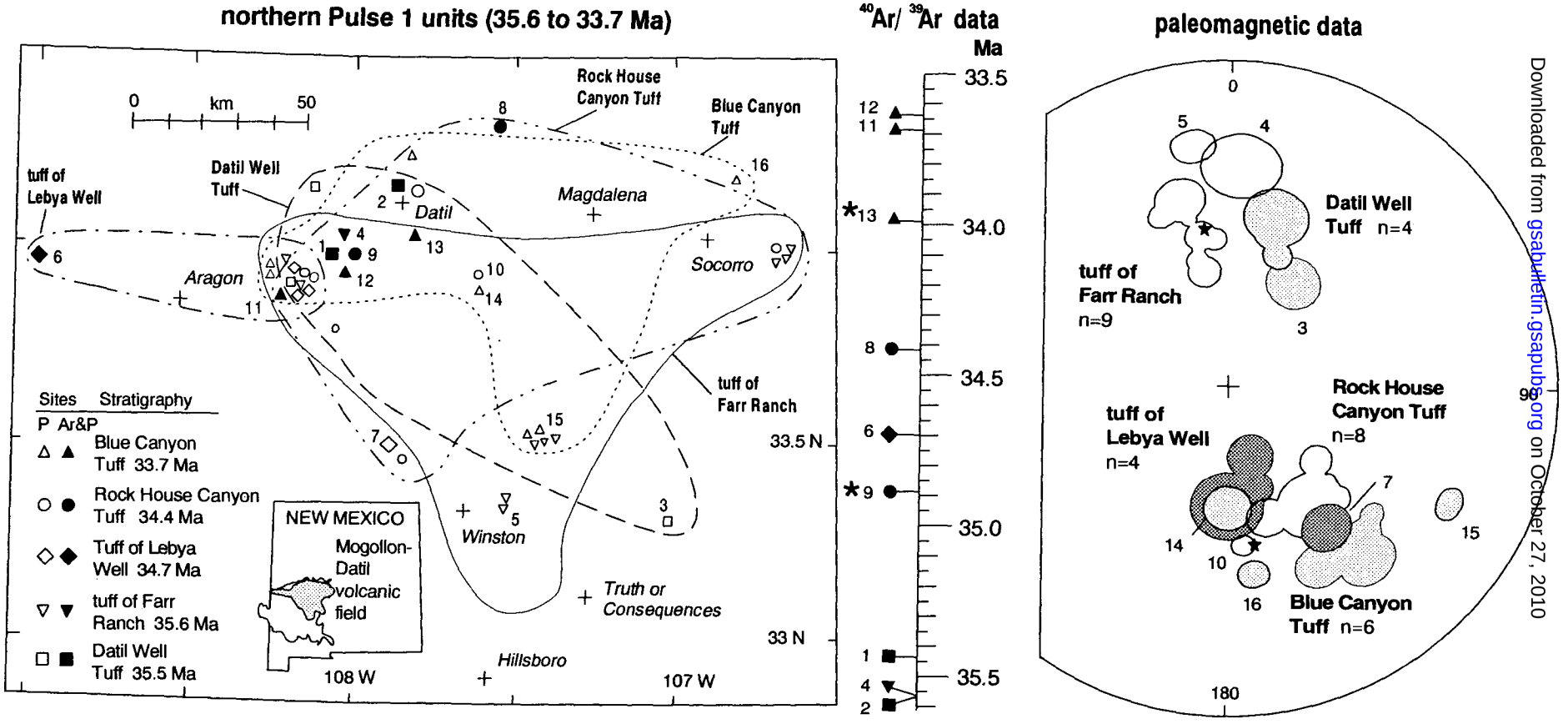


Figure 6. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 35.6 to 33.7 Ma pulse 1 ignimbrites in the northern Mogollan-Datil volcanic field. Correlations of these thin but widely dispersed units are speculative, and no source cauldrons have been established.

Datil Well Tuff and tuff of Farr Ranch. These lithologically similar ignimbrites can be readily distinguished from each other using paleomagnetic data; however, suggested correlations with outcrops to the south (including sites 5 and 3) are only conjectural.

Tuff of Lebya Well. Steep paleomagnetic directions support correlation with 34.70 Ma tuff of Bishop Peak (site 6) but oppose correlation with a lithologically similar unit to the south (site 7).

Rock House Canyon and Blue Canyon Tuffs. One sample from each unit yielded anomalous $^{40}Ar/^{39}Ar$ age data (sites 9 and 13, marked by asterisks), probably reflecting contamination by older feldspar. Paleomagnetic data support tentative correlation of these ignimbrites with units exposed north of Winston. Symbols are explained in Figure 5 caption. Site 13 lacks paleomagnetic data.

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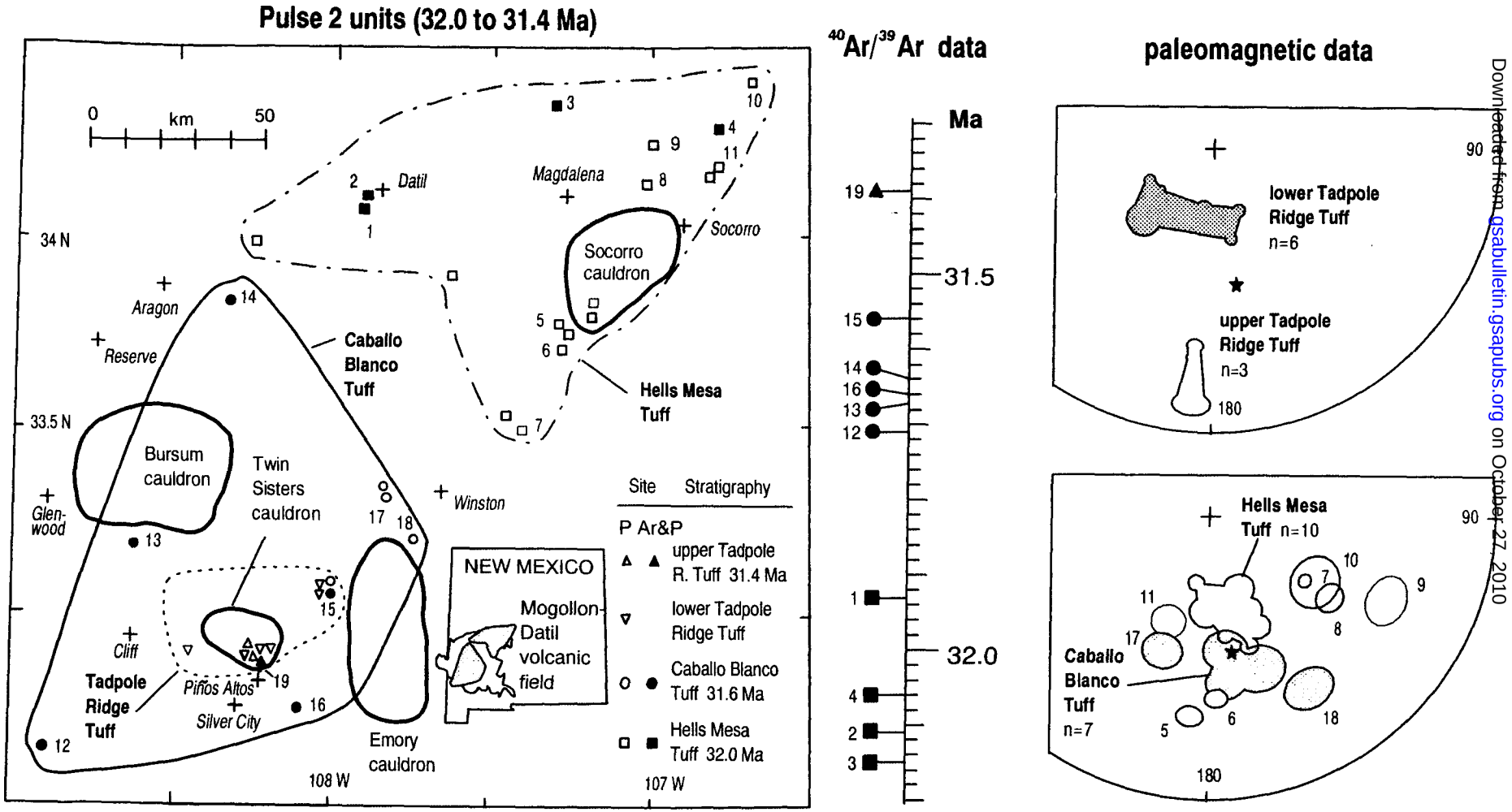


Figure 7. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 32.0 to 31.6 Ma pulse 2 ignimbrites.

Hells Mesa Tuff. The large fraction of anomalous sites reflects sampling in tectonically extended and faulted areas. The Socorro cauldron is well established as the source of Hells Mesa Tuff (Osburn and Chapin, 1983a).

Caballo Blanco Tuff. Correlations based on $^{40}Ar/^{39}Ar$ data indicate that this outflow sheet is larger than previously believed. Its source is unknown, but possibilities include Bursum, Twin Sisters, and Emory cauldrons.

Tadpole Ridge Tuff. The relatively large difference in paleomagnetic direction between upper and lower members is evidence for a substantial time gap between their eruptions. Steep site-mean RM directions in the outflow facies support correlation with the lower intracauldron member. Symbols are explained in Figure 5 caption. Site 1 lacks paleomagnetic data.

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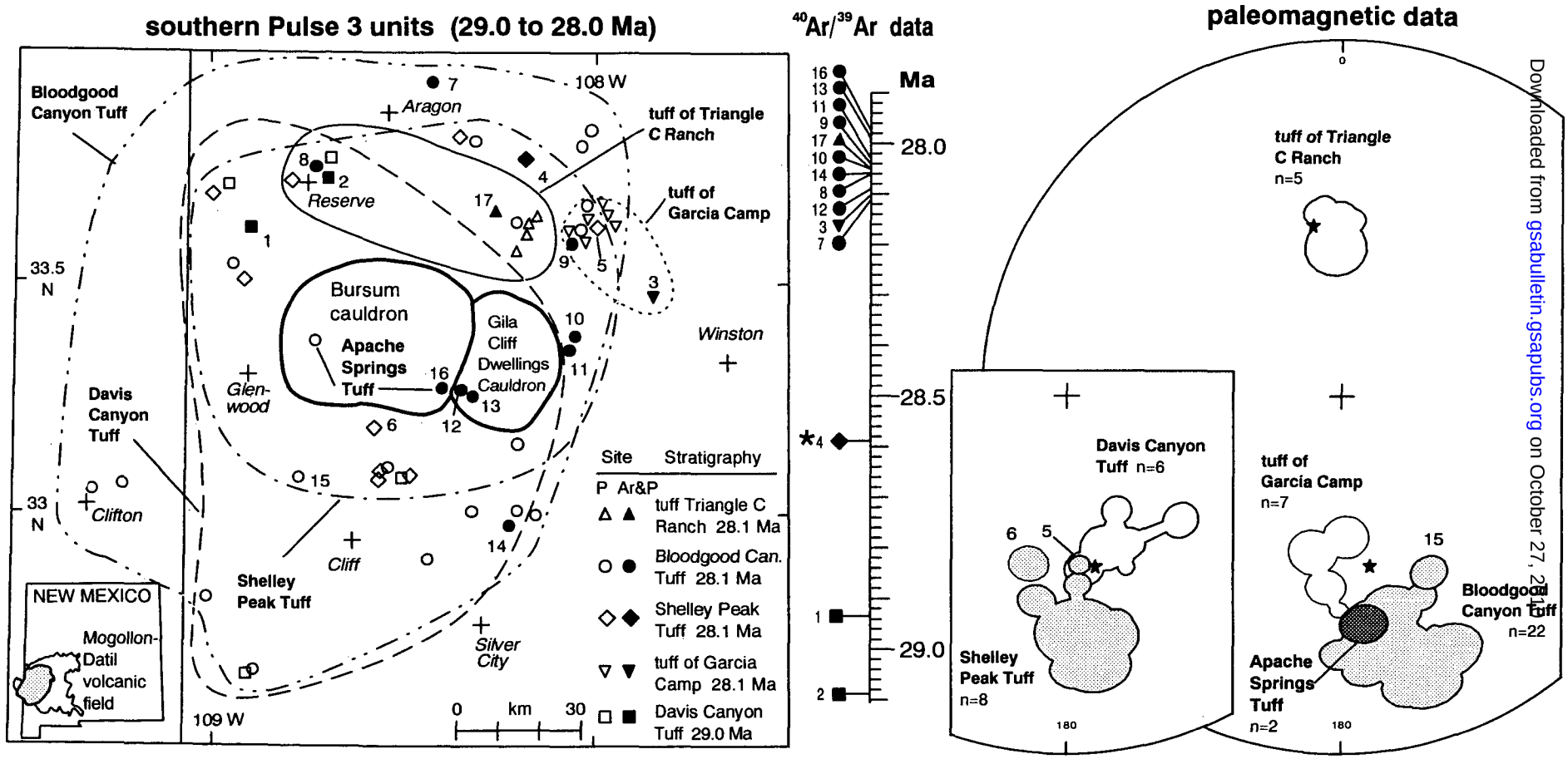


Figure 8. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 29.0 to 28.0 Ma pulse 3 units in the southern Mogollon-Datil field.

Davis Canyon Tuff. Paleomagnetic directions are well grouped over the outflow sheet extent indicated by Ratté and others (1984). Gila Cliff Dwellings cauldron is a possible source for the unit.

Tuff of Garcia Camp. This multi-cooling-unit tuff is part of a pyroclastic apron surrounding Indian Peak rhyolitic domes (Duffield and others, 1988). Paleomagnetic data are consistent with rapid eruption of the entire sequence of cooling units.

Shelley Peak Tuff. The only dated sample of this sanidine-poor outflow sheet has an anomalously old age (site 4, starred), probably reflecting contamination by older feldspar. Gila Cliff Dwellings cauldron is a possible source for this ignimbrite (Ratté and others, 1984).

Bloodgood Canyon Tuff. Paleomagnetic and $^{40}Ar/^{39}Ar$ data support the Ratté and others (1984) interpretation of this unit as the most widespread ignimbrite in the Mogollon-Datil volcanic field. Paleomagnetic directions and one $^{40}Ar/^{39}Ar$ age (site 16) also support interpretation of Apache Springs Tuff as the intracauldron equivalent of the outflow-facies *Bloodgood Canyon Tuff*.

Tuff of Triangle C Ranch. Probably a late-stage product of the Bursum cauldron, this locally distributed ignimbrite is similar in age and lithology to Bloodgood Canyon Tuff but has opposite paleomagnetic polarity. Symbols are explained in Figure 5 caption. Sites 11, 12, and 13 lack paleomagnetic data.

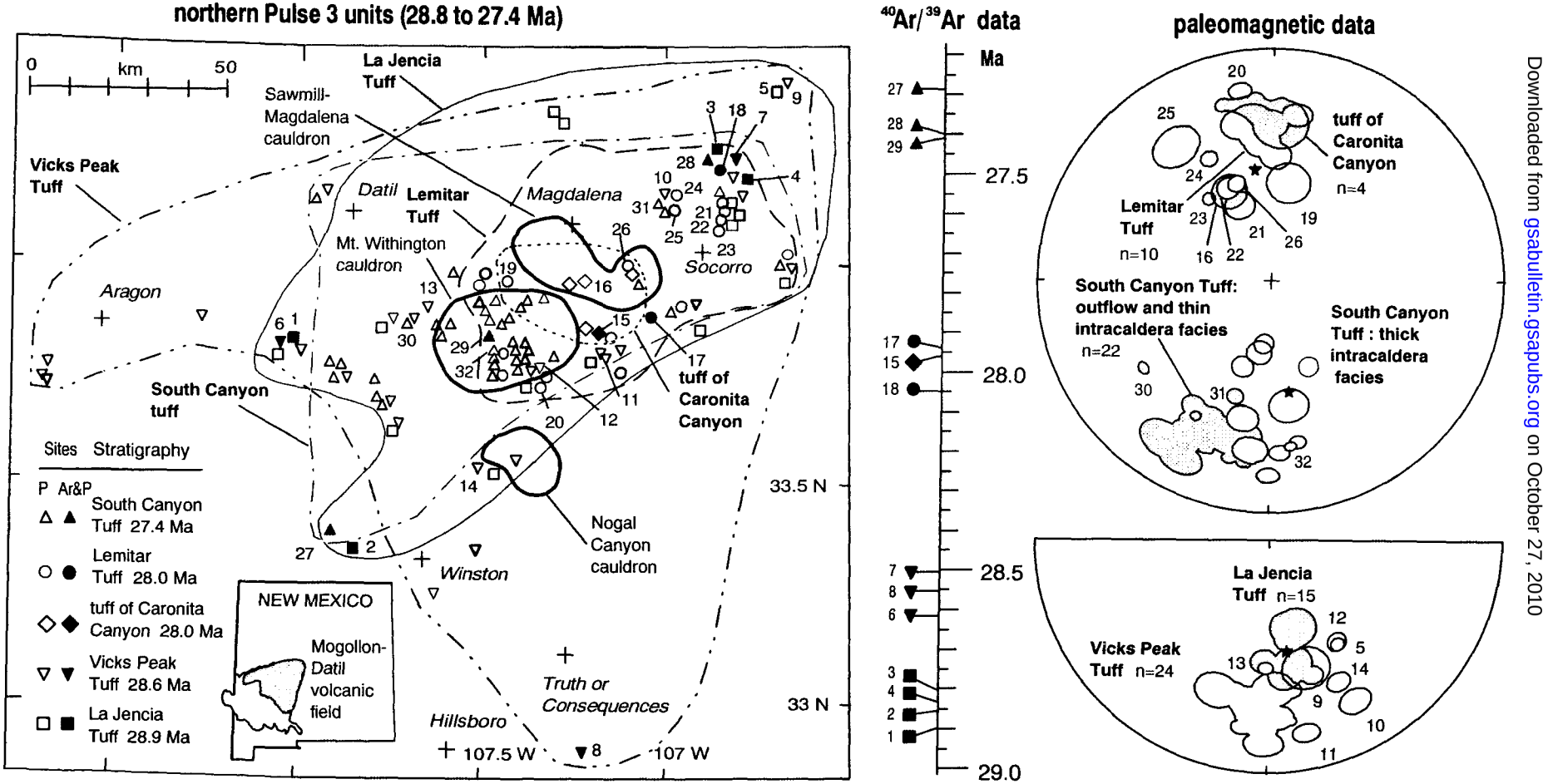


Figure 9. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 28.8 to 27.4 Ma ignimbrites in the northern Mogollon-Datil volcanic field.

La Jencia Tuff. Sawmill-Magdalena cauldron is the well-established source of this unit.

Vicks Peak Tuff. Paleomagnetic and $^{40}Ar/^{39}Ar$ data indicate that this widespread outflow sheet extends farther south and east than previously known; it correlates with Bell Top tuff 7 of Clemons (1976) and parts of tuff of Tularosa Canyon of Rhodes and Smith (1976). Nogal Canyon cauldron is well established as the source.

Caronita Canyon and Lemitar Tuff. Data indicate that these two ignimbrites were erupted in rapid succession; their source is not well established but probably was coincident with the eastern part of the Mount Withington cauldron.

South Canyon Tuff. Mount Withington cauldron is the source of this ignimbrite. Paleomagnetic directions in the outflow and shallow intracauldron facies (shaded) are well grouped, but sites in thicker (>500 m) intracauldron-facies sites (unshaded) have large between-site scatter, probably due to secular variation during protracted cooling (McIntosh, 1991). Symbols are explained in Figure 5 caption. Sites 3, 4, 18, 19, and 28 lack paleogmanetic data.

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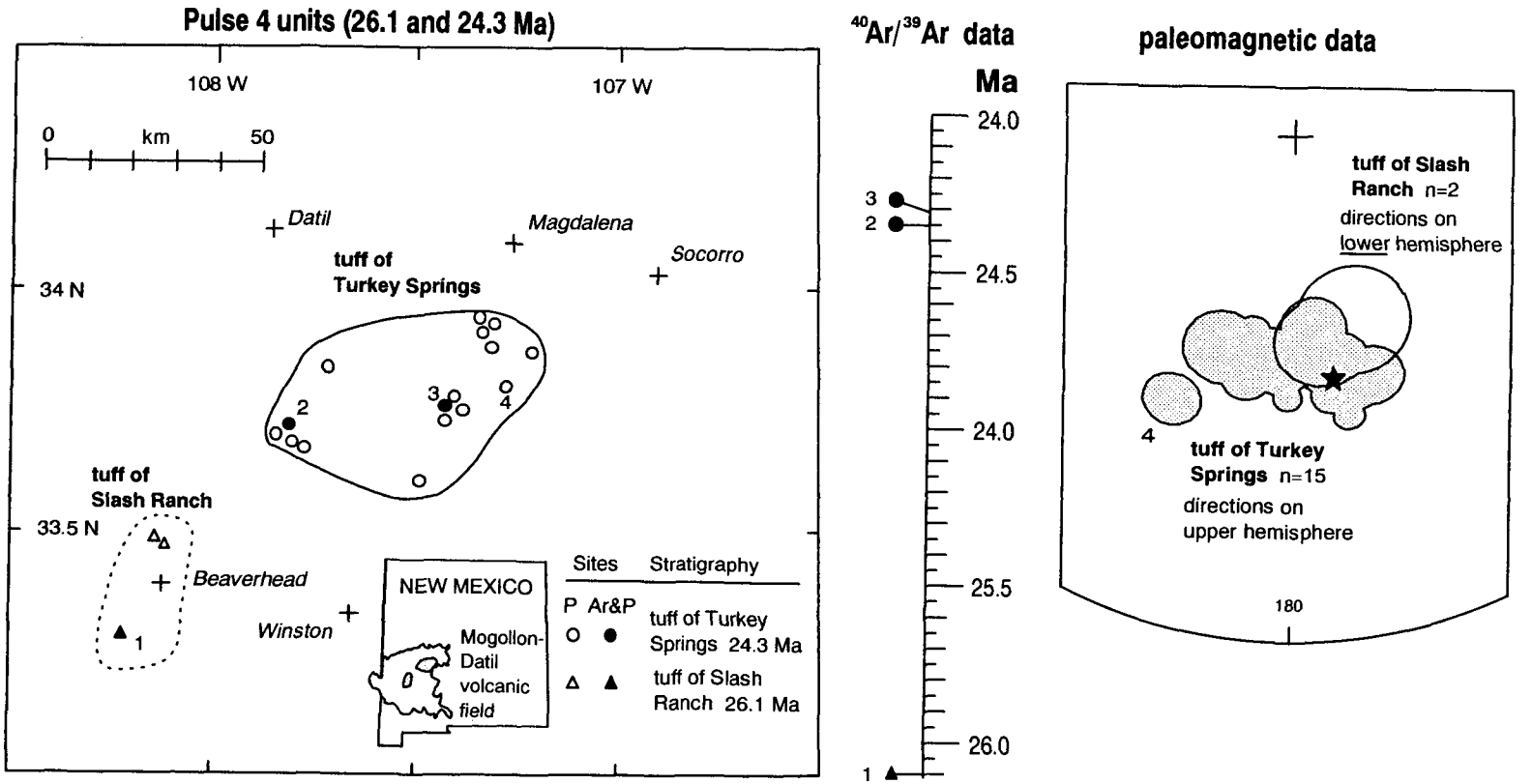


Figure 10. Site locations, areal extent, $^{40}Ar/^{39}Ar$ plateau ages, and site-mean paleomagnetic data for 26.1 and 24.3 pulse 4 ignimbrites.

Slash Ranch. This relatively local, multi-flow ignimbrite is probably related to rhyolitic dome emplacement. Sites from this unit have unusual southward, steeply downward paleomagnetic directions which may reflect emplacement during a geomagnetic field excursion.

Tuff of Turkey Springs. Paleomagnetic and $^{40}Ar/^{39}Ar$ data allow this unit to be readily distinguished from the lithologically similar South Canyon Tuff. The source cauldron has not yet been established but probably lies near the center of the relatively small outflow sheet. Symbols are explained in Figure 5 caption. Site 1 lacks paleomagnetic data.

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nearby Doña Ana Mountains expose an intracauldron sequence that includes a >800-m-thick ignimbrite (Doña Ana Rhyolite Tuff) which is mineralogically distinct from the Organ Mountain units (Seager and others, 1976). $^{40}Ar/^{39}Ar$ plateau ages from single samples of the Cueva, Squaw Mountain, and Doña Ana Tuffs are 36.2, 35.8, and 35.5 Ma, respectively (Table 1). Although based on few samples, some of which have low radiogenic yields probably related to slight alteration, these data are consistent with established stratigraphy. The Cueva Tuff has reversed paleomagnetic polarity, and the Achenback Park, Squaw Mountain, and Doña Ana Tuffs have normal polarity. Site-mean directions are considerably scattered within units (Fig. 5), but such scatter is typical for other thick (>500 m) intracauldron-facies ignimbrites in the Mogollon-Datil volcanic field (McIntosh, 1989, 1991) and elsewhere (for example, Reynolds and others, 1986). Taken together, $^{40}Ar/^{39}Ar$ ages and paleomagnetic directions are consistent with an interpretation of the Organ and Doña Ana sequences as products of a single cauldron complex that was active over a 36.2- to 35.5-Ma interval, during which a reverse-to-normal geomagnetic polarity transition occurred.

West and north of the Organ/Doña Ana intracauldron-facies exposures, throughout the southeastern corner of the volcanic field, there is a generally poorly defined sequence of several thin (<50 m), commonly poorly welded, outflow sheets (Table 1), including tuffs 2, 3, and 4 of the Bell Top Formation (Clemons, 1976) and ignimbrite members of the Sugarlump Tuff (Elston, 1957; Seager and others, 1982). $^{40}Ar/^{39}Ar$ ages obtained from these units range from 35.7 to 34.9 Ma, and all sites have normal magnetic polarity (Table 1 and Fig. 5). One or more of the lower units may represent outflow facies erupted from the Organ/Doña Ana cauldron. Tuff 3 of the Bell Top Formation (Clemons, 1976) is the best candidate for a co-eruptive outflow sheet for the Achenback Park/Squaw Mountain sequence because of its similar age (35.7 Ma), normal magnetic polarity, and geographic proximity (Fig. 5). A second possibility is the underlying tuff 2 of the Bell Top Formation (Clemons, 1976), which has not yet been dated by the $^{40}Ar/^{39}Ar$ method. Correlations remain uncertain between these units and similar-age units farther west (Sugarlump Tuff and other units presented in McIntosh and others, 1991).

More confident correlations can be made for Bell Top Formation tuff 4 (Clemons, 1976), the youngest pre–Kneeling Nun regional ignimbrite. Similar northward, steep paleomagnetic directions and $^{40}Ar/^{39}Ar$ ages near 34.9 Ma support correlation of Bell Top Formation tuff 4 in the Goodsight–Cedar Hills area with a widespread, lithologically similar, pre–Kneeling Nun unit in the Winston area (tuff of Rocque Ramos Canyon of Harrison, 1990) and the Salado Mountains (unnamed crystal tuff of Seager and Mayer, 1988). Geographic proximity to the Emory cauldron suggests that this 34.9 Ma outflow sheet may represent a precursor unit derived from the same magma chamber as 34.9 Ma Kneeling Nun Tuff. In the Winston area, a thin but paleomagnetically distinct outflow sheet (tuff of Stone House Ranch, Fig. 5 and Table 1) is present between the Bell Top tuff 4 correlative and the Kneeling Nun Tuff; this unit may represent an additional precursor to eruption of Kneeling Nun Tuff.

Two pre–Kneeling Nun regional ignimbrites are also present in the northern part of the volcanic field (Fig. 6). Datil Well Tuff (35.5 Ma), a well-exposed unit in the Datil area, has a distinctive northeastward TRM direction (Fig. 6). Datil Well Tuff may correlate with a lithologically and paleomagnetically similar unit exposed in isolated outcrops 125 km southeast of the Datil area (site 3, Fig. 6). In the area southwest of Datil (Fig. 6), Datil Well Tuff is overlain by the lithologically similar but paleomagnetically distinct tuff of Farr Ranch, which has yielded a statistically indistinguishable $^{40}Ar/^{39}Ar$ age of 35.6 Ma (Table 1 and Fig. 6). Paleomagnetic directions indicate that some outcrops southeast of Socorro, formerly mapped as Datil Well Tuff (Osburn and Chapin, 1983b), are actually tuff of Farr Ranch (Fig. 6). The tuff of Farr Ranch may also correlate with two very poorly known units exposed 40 to 100 km to the south (tuff of Luna Park of Hermann, 1986; tuff of Victoria Tank of Harrison, 1990). Source cauldrons have not been established for either Datil Well or Farr Ranch outflows sheets, but their geographic distributions are consistent with eruption from vents somewhere in the northern part of the field.

Kneeling Nun Tuff (34.9 Ma). The volcanic field's oldest documented large-volume ignimbrite is the relatively well-studied Kneeling Nun Tuff, erupted from the Emory cauldron in the southern Black Range (Elston and others 1975). $^{40}Ar/^{39}Ar$ ages, which cluster near 34.9 Ma, serve to readily distinguish Kneeling Nun Tuff from lithologically similar younger units (Box Canyon, Hells Mesa, and Caballo Blanco; Table 1 and Fig. 5). In addition, $^{40}Ar/^{39}Ar$ data support correlation of the Kneeling Nun Tuff with ignimbrites formerly thought to be local units, including Bell Top Formation tuff 5 in the Goodsight–Cedar Hills area (Clemons, 1976), part of the mapped Sugarlump Tuff near Faywood (Elston, 1957; Seager and others, 1982), and a thin, unwelded unit overlying the tuff of Farr Ranch in the area southwest of Datil (Ratté and others, 1989).

Paleomagnetic data are only marginally helpful for addressing Kneeling Nun correlation questions. Kneeling Nun Tuff and lithologically similar Hells Mesa, Caballo Blanco, and Box Canyon Tuffs all have non-distinctive unit-mean directions similar to the Eocene-Oligocene reference direction (Figs. 4, 5, and 7 and Table 1). Furthermore, these four units have generally large within-unit scatter of site-mean paleomagnetic directions and a relatively large proportion of paleomagnetically discordant sites (Figs. 4, 5, and 7), probably mainly reflecting errors in determining the paleohorizontal in these crystal-rich, pumice-poor ignimbrites (McIntosh, 1991). Elston and others (1975) estimated an eruptive volume of 900 km^3 for the Kneeling Nun Tuff; the above-mentioned correlations indicate that the eruptive volume may actually have been significantly larger.

Post–Kneeling Nun, Pre–Box Canyon Units. During the 1.3-m.y. interval following eruption of the Kneeling Nun Tuff, the southern part of the Mogollon-Datil volcanic field was relatively quiescent. Mimbres Peak rhyolite domes and associated local ignimbrites were erupted near 34.6 Ma (Table 1) in the southern Black Range, at least in part as moat-fill of the Emory cauldron (Elston and others, 1975). Locally exposed units along the southern edge of the field, including the 34.0 Ma tuff of Table Mountain (Table 1), probably represent northern fringes of units erupted from the "boot heel" area south of the Mogollon-Datil volcanic field. Another unit that probably was erupted during this interval is the Cooney Tuff, an ignimbrite exposed near the western edge of the field (Fig. 5). In places, the Cooney Tuff consists of several thin ignimbrite cooling units separated by sedimentary sequences (Ratté and others, 1984). Published K-Ar biotite ages from the Cooney Tuff average 34 Ma (Marvin and others, 1991), but samples bearing fresh sanidine suitable for $^{40}Ar/^{39}Ar$ dating have not been found. Three sites in outcrops mapped as Cooney Tuff have dispersed site-mean TRM directions (Fig. 5), suggesting that eruption of the Cooney Tuff members spanned an interval that was long relative to rates of secular variation—at the very least, many centuries. Long-range paleomagnetic correlation of this unit, therefore, appears to be impractical.

During this interval of relative quiescence in the southern part of the field, eruptions in the north produced three regional ignimbrites: the tuff of Lebya Well (34.7 Ma), Rock House Canyon Tuff (34.4 Ma), and Blue Canyon Tuff (33.7 Ma). The tuff of Lebya Well was first recognized in the area southwest of Datil by its stratigraphic position between 34.9 Ma Kneeling Nun and 34.4 Ma Rock House Canyon Tuffs (Fig. 6; Ratté and others, 1989). Although the tuff of Lebya Well has not been directly dated in this area, it may correlate with the 34.7 Ma Bishop Peak Tuff, exposed 75 km to the west, which has a distinctively steep TRM direction similar to that of the tuff of Lebya Well (Fig. 6). Paleomagnetic directions and $^{40}Ar/^{39}Ar$ ages from the Rock House Canyon and Blue Canyon Tuffs generally confirm the mapped extent of these lithologically distinctive ignimbrites (Osburn and Chapin, 1983a) and also support correlations with isolated outcrops farther south (Fig. 6). Like Datil Well Tuff and tuff

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of Farr Ranch, source cauldrons are unknown for the Lebya Well, Rock House Canyon, and Blue Canyon outflow sheets, although their distributions are consistent with vent locations in the northern part of the field.

Box Canyon Tuff (33.5 Ma). Box Canyon Tuff is the final ignimbrite erupted in Episode 1. This unit was first identified by Elston (1957) from its stratigraphic position between Kneeling Nun and Caballo Blanco Tuffs, but it has not been previously recognized as a regional ignimbrite. $^{40}Ar/^{39}Ar$ plateau ages near 33.5 Ma, together with somewhat steep TRM directions (Fig. 6), support correlation of Box Canyon Tuff with several locally mapped units distributed over much of the southern and western edges of the volcanic field (Fig. 5). These include Bell Top 6 tuff (Clemons, 1976), parts of mapped Sugarlump Tuff (Elston, 1957), mapped Kneeling Nun Tuff of Hedlund (1978), Cherokee Canyon Tuff (Finnell, 1987), parts of Fall Canyon Tuff (Ratté and others, 1984), and tuff of Luna (T. L. Finnell, 1988, personal commun.). A probable source for the tentatively correlated Box Canyon Tuff outflow sheet is the poorly known Schoolhouse Mountain cauldron, which contains an intracauldron-facies ignimbrite yielding an $^{40}Ar/^{39}Ar$ age of 33.65 Ma (Fig. 5).

Pulse 2 (32.1 to 31.4 Ma): A Short Burst

The second pulse of Mogollon-Datil ignimbrite activity (Fig. 7) spanned a 0.7-m.y. interval beginning at 32.1 Ma, 1.5 m.y. after eruption of the Box Canyon Tuff. Three regional ignimbrite outflow sheets were produced: Hells Mesa Tuff (32.1 Ma) in the northern part of the field, and Caballo Blanco Tuff (31.6 Ma) and Tadpole Ridge Tuff (31.4 Ma) in the southern part of the field. As discussed below, source cauldrons have been established only for Hells Mesa and Tadpole Ridge Tuffs.

Hells Mesa Tuff (32.1 Ma). The 1,250-km^3 Hells Mesa Tuff erupted at 32.1 Ma from the Socorro cauldron (Fig. 7). This well-studied unit (Osburn and Chapin, 1983a, 1983b) is the oldest of five large-volume regional ignimbrites exposed in the northern field. The Hells Mesa outflow sheet has a highly asymmetric distribution around the cauldron, extending almost 100 km to the west but absent to the south and east (Fig. 7). Paleomagnetic directions and $^{40}Ar/^{39}Ar$ ages are consistent with the previously mapped extent of the Hells Mesa Tuff outflow sheet and further support correlation of the Hells Mesa Tuff with distal outflow-facies ignimbrites exposed in the southern San Mateo Mountains and the area southwest of Datil (Fig. 7). $^{40}Ar/^{39}Ar$ age determinations from the Hells Mesa Tuff (32.06 ± 13 Ma; n = 4) provide a strong correlation criterion because they can be clearly resolved from all lithologically similar units, including the 31.6 Ma Caballo Blanco Tuff (Fig. 7). Paleomagnetic data are somewhat less useful, in part because of the non-distinctive unit-mean direction (Fig. 4) (only 11° from the Eocene-Oligocene reference direction), and in part because of the high fraction (7 out of 17) of sites which have discordant mean directions (Fig. 7). As detailed in McIntosh (1989, 1991), most of these discordant directions are attributable to fault-related rotations in strongly extended areas in the northeastern part of the field.

Caballo Blanco Tuff (31.7 Ma) and Tadpole Ridge Tuff (31.4 Ma). Shortly after eruption of the Hells Mesa Tuff in the northern Mogollon-Datil volcanic field, the Caballo Blanco Tuff (31.7 Ma) was erupted in the southern part of the field. Although recognized early as a distinct ignimbrite (Elston, 1957; Elston and others, 1973), correlation of the Caballo Blanco Tuff has been hampered by lithologic similarity to the Kneeling Nun, Box Canyon, and Hells Mesa Tuffs. The Caballo Blanco Tuff can now be reliably distinguished from these units using its $^{40}Ar/^{39}Ar$ age (31.65 ± .06 Ma; n = 5). The Caballo Blanco outflow sheet is now known to extend at least 70 km west and 100 km north of its original mapped extent in the Black Range and Silver City areas (Fig. 7; Elston, 1976), correlating with part of the Fall Canyon Tuff mapped by Ratté and others (1984), including the Fall Canyon Tuff at its type locality (site 13, Fig. 7). Like the case of the Hells Mesa Tuff, paleomagnetic data are not particularly distinctive but are consistent with a wide distribution for the unit (Fig. 7).

Although the source cauldron for the Caballo Blanco Tuff has not been established, three possibilities are under consideration. These are (1) the southern end of the Emory cauldron, as suggested in Abbitz (1989); (2) a now obliterated precursor to the Bursum cauldron, which is consistent with abundant large lithic fragments in the Caballo Blanco Tuff near the margin of the Bursum cauldron (site 13, Fig. 7); and (3) an as yet unrecognized cauldron coincident with the younger Twin Sisters cauldron, which is near the center of the distribution of the Caballo Blanco outflow sheet (Fig. 7).

Eruption of the Tadpole Ridge Tuff (31.4 Ma) followed soon after emplacement of the Caballo Blanco Tuff. This geographically restricted, mineralogically distinctive ignimbrite (low-silica rhyolite, 10%–25% plagioclase and biotite phenocrysts) was originally mapped as a local unit by Finnell (1976), who recognized distinct upper and lower members. Much of the mapped Tadpole Ridge Tuff appears to lie within a small cauldron (herein termed the "Twin Sisters cauldron") which has a southern margin near Piños Altos (Fig. 7) and a northern margin buried by younger units. Evidence for the southern cauldron margin includes abrupt thickness changes, coarse lithic breccias, and a high-angle basal unconformity.

Paleomagnetic data indicate a time break between eruption of the upper and lower members of the Tadpole Ridge Tuff. Although neither paleosols nor substantial sedimentary deposits are found along their mutual contact, the lower and upper intracauldron-facies members, respectively, have unusually steep and unusually shallow unit-mean TRM directions that are 49° apart (Fig. 7). This difference in TRM directions probably reflects an interval of at very least 1,000 yr between eruption of the two members, given a typical average secular variation rate of 4.5° per century (Holcomb and others, 1986). Alternatively, the difference in TRM directions may have resulted from two closely spaced eruptions that occurred during a rapidly changing field excursion or polarity transition event. Distinctively steep site-mean paleomagnetic data (Fig. 7) further provide a strong criterion for correlation of lower member Tadpole Ridge Tuff with adjacent thin, plagioclase-biotite, outflow-facies units mapped as tuff of Terry Canyon by Krier (1981) and as an unnamed local unit by Finnell (1982).

Due to low sanidine content (<0.5%), it has proven difficult to obtain an accurate $^{40}Ar/^{39}Ar$ age for the Tadpole Ridge Tuff. An incorrect published age of 35.14 Ma (McIntosh and others, 1986) probably represents extreme xenocrystic contamination primarily from the 34.9 Ma Kneeling Nun Tuff, which is exposed in the nearby cauldron wall and locally forms abundant lithic fragments. The present 31.4 Ma age for a single sample of the upper member of the Tadpole Ridge Tuff agrees closely with published K-Ar ages (31.6 and 31.9 Ma, Marvin and others, 1987) but should be considered tentative until more data are obtained.

Pulse 3 (29.0 to 27.4 Ma): The Big Doublet

The third and largest pulse of Mogollon-Datil ignimbrite volcanism produced nine major regional ignimbrites during a 1.6-m.y. interval from 29.0 to 27.4 Ma. Pulse 3 can actually be resolved into a closely spaced pair of shorter pulses: pulse 3A produced crystal-poor ignimbrites between 29.0 and 28.5 Ma, and pulse 3B produced relatively crystal-rich ignimbrites between 28.0 and 27.4 Ma. During each of these shorter pulses, two separate clusters of cauldrons were active: one in the Mogollon Mountains in the southwestern corner of the field, and the second west of Socorro in the northern end of the field (Figs. 2, 8, and 9).

Southern Pulse 3A Units: Local Black Range Ignimbrites (29.1 to 28.6 Ma) and Davis Canyon Tuff (29.0 Ma). The oldest pulse 3 units are local, rhyolite-dome–related ignimbrites exposed in the central Black Range (tuffs of Monument Canyon, Little Mineral Creek, Stiver Canyon,

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and Mud Hole; Table 1). Although these units cannot be traced over wide areas, their $^{40}Ar/^{39}Ar$ ages range from 29.1 to 28.6 Ma, and paleomagnetic data indicate that their eruption spanned a normal-to-reversed polarity reversal near 29.0 Ma (Table 1).

The first major regional ignimbrite erupted in pulse 3 is the >10,500 km^2 Davis Canyon Tuff (Fig. 8) exposed in the southwestern part of the volcanic field. Two $^{40}Ar/^{39}Ar$ plateau ages averaging 29.0 Ma were obtained from the northern edge of this crystal-poor, high-silica rhyolitic outflow sheet, and site-mean TRM direction from six sites are well grouped, supporting the distribution mapped by Ratté and others (1984). Paleomagnetic data (Fig. 8) are useful for distinguishing this unit from the lithologically similar Vicks Peak Tuff, which has lower paleomagnetic inclination (Fig. 9). This distinction is particularly important in the overlap area of the two ignimbrites, where both were formerly mapped as tuff of Tularosa Canyon (Rhodes and Smith, 1976). The source cauldron for the Davis Canyon Tuff is not known, although its distribution is consistent with a source in either the Bursum or Gila Cliff Dwellings cauldron areas.

Northern Pulse 3A Units: La Jencia Tuff (28.9 Ma) and Vicks Peak Tuff (28.5 Ma). Two crystal-poor, high-silica rhyolitic ignimbrites, the La Jencia Tuff and Vicks Peak Tuff, were erupted from the northern cauldron complex during the early phase of pulse 3. These ignimbrites have distinct $^{40}Ar/^{39}Ar$ ages and paleomagnetic directions (Fig. 9), which assist in distinguishing one unit from the other and in identifying their distal fringes, particularly in areas of potential overlap with Davis Canyon Tuff.

Paleomagnetic directions and $^{40}Ar/^{39}Ar$ data from the La Jencia Tuff are consistent within its previously mapped extent, and furthermore indicate that the unit extends even farther to the southwest, as much as 80 km from its cauldron margin. Identified distal fringes of the La Jencia Tuff include outcrops along the south edge of the Plains of San Agustin (site 1, Fig. 9) and in the central Black Range (site 2, Fig. 9). The source of the La Jencia Tuff is well established as the Sawmill-Magdalena cauldron (Fig. 9; Osburn and Chapin, 1983b).

$^{40}Ar/^{39}Ar$ ages and distinctively shallow paleomagnetic directions have helped to demonstrate that the Vicks Peak Tuff is the most widely distributed ignimbrite in the northern Mogollon-Datil volcanic field (Fig. 9). Distal correlatives include tuff 7 of the Bell Top Formation (Clemons, 1976) and the upper member of the tuff of Tularosa Canyon (Rhodes and Smith, 1976). Several paleomagnetic sites in Vicks Peak Tuff have discordant declinations, primarily due to fault-block rotations in highly extended areas (Fig. 9; McIntosh, 1991). The source of Vicks Peak Tuff was the Nogal Canyon cauldron (Fig. 9; Osburn and Chapin, 1983b).

Southern Pulse 3B Units: Local Ignimbrites, Shelley Peak Tuff (28.1 Ma), Bloodgood Canyon Tuff (28.1 Ma), and Tuff of Triangle C (28.0 Ma). The stratigraphically oldest pulse 3B units are the Taylor Creek Rhyolite sequence of domes and related pyroclastic rocks (Duffield and others, 1987); Dalrymple and Duffield, 1988), erupted in the northern Black Range in the area between the Mogollon and Socorro cauldron clusters (Fig. 2). One subregional rhyolite-dome–derived ignimbrite, the tuff of Garcia Camp, was studied in some detail (Fig. 8). Paleomagnetic site-mean directions from the tuff of Garcia Camp are tightly grouped (Fig. 8), indicating that this multi-cooling-unit ignimbrite was erupted over a brief interval. A single $^{40}Ar/^{39}Ar$ plateau age of 28.10 Ma was determined, which compares to a published mean $^{40}Ar/^{39}Ar$ laser-fusion age of 28.21 ± .04 Ma for sanidine from other samples of the tuff of Garcia Camp and co-eruptive rhyolitic lavas (Dalrymple and Duffield, 1988). The interlaboratory difference in ages is discussed in McIntosh and others (1990).

Overlying the 28.1 Ma tuff of Garcia Camp is the Shelley Peak Tuff, a widespread, red-colored, plagioclase-biotite–rich, low-silica rhyolitic ignimbrite erupted from a source in the Mogollon cauldron cluster, possibly the Gila Cliff Dwellings cauldron (Ratté and others, 1984). Lithology is the most reliable available correlation criterion for the Shelley Peak Tuff. The unit contains abundant distinctive green pyroxene phenocrysts which allow it to be distinguished from the otherwise lithologically similar Tadpole Ridge and Apache Springs Tuffs. Paleomagnetic data from this unit are moderately well grouped in a shallow-inclination, southward direction, although some of the sites have slightly discordant directions and demagnetization behavior indicative of minor chemical remanent magnetization (CRM) components in addition to TRM (Fig. 8; McIntosh, 1989, 1991). The TRM directions are sufficient to differentiate Shelley Peak Tuff from the Tadpole Ridge Tuff (Fig. 7), but not from the paleomagnetically similar Apache Springs Tuff (Fig. 8). A single $^{40}Ar/^{39}Ar$ plateau age determined for the Shelley Peak Tuff is 28.52 ± .08 Ma. This anomalously old age probably reflects contamination of the sanidine-poor Shelley Peak Tuff by older lithic or xenocrystic feldspar (McIntosh and others, 1990); however, because the Shelley Peak Tuff is stratigraphically bracketed between the underlying tuff of Garcia Camp (28.1 Ma) and overlying Bloodgood Canyon Tuff (28.1 Ma), its age is determined as precisely as if it were itself well dated.

Overlying the Shelley Peak Tuff, the 15,000-km^2, 1,000-km^3 Bloodgood Canyon Tuff is found; it is the most widely exposed ignimbrite in the southern part of the volcanic field (Ratté and others, 1984). The Bloodgood Canyon Tuff was almost certainly erupted from the 50-km-wide Bursum cauldron (Ratté and others, 1984), although other sources have been suggested (Elston, 1984). Paleomagnetic data from throughout the extent of this high-silica rhyolitic outflow sheet have distinctive, shallow-inclination, southeastward TRM directions (Fig. 8). Similarly, $^{40}Ar/^{39}Ar$ plateau ages from seven widely separated samples are nearly identical, ranging from 28.11 to 27.99 Ma and averaging 28.05 ± .04 Ma (Fig. 8). These paleomagnetic and $^{40}Ar/^{39}Ar$ data strongly support the conclusion that the Bloodgood Canyon Tuff is a single large outflow sheet (Ratté and others, 1984) and oppose the interpretation that exposures north of the Bursum cauldron represent a different eruptive unit, termed the Railroad Canyon Tuff (Elston and others, 1973; Rhodes and Smith, 1976; Elston, 1984). Furthermore, the TRM direction of the Bloodgood Canyon Tuff is sufficiently distinctive to reliably distinguish this unit from all other lithologically similar ignimbrites in the northern part of the field (Figs. 4, 8, 9, and 10).

Within the Bursum cauldron, the 1,200-km^3 Apache Springs Tuff is found (Ratté and others, 1984). $^{40}Ar/^{39}Ar$ and paleomagnetic data are consistent with the interpretation that this unit is the intracauldron-facies equivalent of the Bloodgood Canyon Tuff (Ratté and others, 1984). A single $^{40}Ar/^{39}Ar$ plateau age (27.98 Ma) for the Apache Springs Tuff is statistically indistinguishable from the mean age of Bloodgood Canyon Tuff samples, and site-mean TRM directions from two Apache Springs sites are within 10° of unit-mean direction of the Bloodgood Canyon Tuff.

The northern part of the Bloodgood Canyon Tuff is locally overlain by the poorly welded, but lithologically similar, tuff of Triangle C Ranch, probably also erupted from the Bursum cauldron. The one $^{40}Ar/^{39}Ar$ plateau age from this unit (28.05 Ma, Table 1) is identical to the mean age of the Bloodgood Canyon Tuff age, but the six site-mean TRM directions from the tuff of Triangle C Ranch are well grouped in a *normal* polarity direction. Apparently, this unit was erupted soon after a polarity reversal that closely followed the eruption of the Bloodgood Canyon Tuff. It follows that these two eruptions were separated by 5,000 yr or more, according to estimates for the duration of geomagnetic polarity reversals. Data from these two ignimbrites provide a good example of how paleomagnetic study can augment the resolution of $^{40}Ar/^{39}Ar$ age determinations.

Northern Pulse 3B Units: Tuff of Caronita Canyon (28.0 Ma), Lemitar Tuff (28.0 Ma), and South Canyon Tuff (27.4 Ma). The Lemitar Tuff is the oldest widespread unit overlying the 28.5 Ma Vicks Peak Tuff in the northern part of the volcanic field. The source cauldron for this well-mapped unit has not been identified, but thickness and distri-

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bution are consistent with a source in the west-central Magdalena Mountains. Evidence for this cauldron has apparently been obscured or obliterated by the 27.4 Ma Mount Withington cauldron (Figs. 2 and 9). Two $^{40}Ar/^{39}Ar$ age determinations from the Lemitar Tuff average 28.0 Ma; this age and normal magnetic polarity serve to readily distinguish the Lemitar Tuff from the lithologically similar, reverse-polarity, 27.4 Ma South Canyon Tuff. About half of the sites in the Lemitar Tuff have tightly grouped site-mean TRM directions, but the remainder are somewhat scattered (Fig. 9). For the most part, these discordant site-mean directions are related to block rotations in the highly extended Lemitar Mountains and Joyita Hills (McIntosh, 1991).

Central parts of the Lemitar Tuff are underlain by the strongly zoned tuff of Caronita Canyon (Fig. 9), formerly interpreted as a part of the moat-fill sequence emplaced after eruption of the 28.8 Ma La Jencia Tuff (Osburn and Chapin, 1983a). Normal polarity and an $^{40}Ar/^{39}Ar$ age of 28.0 Ma (Table 1 and Fig. 9), however, support an interpretation of the tuff of Caronita Canyon as a precursor unit erupted slightly before the Lemitar Tuff.

The 27.4 Ma South Canyon Tuff is the youngest major, widespread ignimbrite in the northeastern Mogollon-Datil volcanic field. This well-mapped, high-silica rhyolite was erupted from the Mount Withington cauldron, a "trapdoor" cauldron hinged along its southern edge (Fig. 9). Outflow and intracauldron facies are well exposed and were extensively sampled (43 sites) for paleomagnetic study. $^{40}Ar/^{39}Ar$ plateau ages from samples of both facies range from 27.28 to 27.40 Ma. Site-mean paleomagnetic data from outflow facies and thinner (<500 m) southern intracaldera facies are well grouped in an unusual, shallow, southwestward direction that is distinct from all other Mogollon-Datil ignimbrites (Figs. 4 and 9). This distinctive TRM direction, together with $^{40}Ar/^{39}Ar$ data, has been used to identify the 1.5-m-thick unwelded distal fringe of the South Canyon Tuff outflow sheet in the central Black Range (site 27, Fig. 9).

Site-mean TRM directions from thicker (500–1,500 m), northern intracauldron facies are scattered (Fig. 9) and have a mean direction which is offset toward the expected time-averaged field direction, relative to the mean direction of the outflow facies (Fig. 9 and Table 1). The preferred explanation for the non-uniform magnetization of the thick intracaldera facies is that it reflects paleosecular variation which occurred during protracted cooling (Reynolds and others 1986; Wells and Hillhouse, 1989). Alternatively, the scattered site-mean directions might reflect nonhorizontality of pumice foliations during initial cooling, or sub-blocking-temperature welding, flowage, or compaction (Rosenbaum, 1986). These possibilities are discussed in detail in McIntosh (1991). In any case, these results indicate that paleomagnetism is less effective as a correlation criterion in thick intracauldron facies than in outflow sheets.

Pulse 4 (26.1 and 24.3 Ma): Last Gasps

Tuff of Slash Ranch (26.1 Ma). The youngest ignimbrite present in the central part of the Mogollon-Datil volcanic field is a poorly welded, zeolitized, but prominently exposed, multi-cooling-unit outflow sheet mapped at its northern end as tuff of Slash Ranch and at its southern end as tuff of Jordan Canyon (Fig. 10). A single $^{40}Ar/^{39}Ar$ age of 26.1 Ma has been determined, and two sites have a highly unusual south and downward remanence direction (Table 1 and Fig. 10). Interpretation of these data is uncertain, however, because of advanced zeolitization and observed nonvolcanic contaminant feldspar. Although no source has been identified, the limited lateral extent (22 km) and multi-cooling-unit nature of this unit suggest that it is related to eruption of a rhyolite dome, rather than being produced by a cauldron-forming eruption.

Tuff of Turkey Springs (24.3 Ma). The youngest ignimbrite outflow sheet in the Mogollon-Datil sequence is the high-silica rhyolite tuff of Turkey Springs, which overlies the South Canyon Tuff in the northern part of the field (Ferguson, 1986). Prior to paleomagnetic and $^{40}Ar/^{39}Ar$ study, this moderately extensive outflow sheet (Fig. 10) was not recognized as a stratigraphically distinct unit but was incorrectly mapped as either South Canyon Tuff (Donze, 1980) or Railroad Canyon Tuff (Fodor, 1976).

Two $^{40}Ar/^{39}Ar$ plateau ages for the tuff of Turkey Springs average 24.3 Ma (Fig. 10 and Table 1), 2.9 m.y. younger than the underlying South Canyon Tuff. Site-mean paleomagnetic directions from this unit cluster tightly around the unit-mean TRM direction. Although the unit-mean TRM direction is only 4° from the time-averaged Oligocene field, it is distinctly different from the unit-mean directions of lithologically similar South Canyon, Lemitar, Bloodgood Canyon, and Triangle C Ranch Tuffs (Figs. 8, 9, and 10). No source has yet been identified for the tuff of Turkey Springs, but the gross distribution of the paleomagnetically identified outflow sheet (Fig. 10) and underlying pyroclastic fall deposit are consistent with a vent area, perhaps a small cauldron, in the east-central San Mateo Mountains, an area which has not yet been mapped in detail.

TIME-STRATIGRAPHIC FRAMEWORK

As detailed in the previous sections, $^{40}Ar/^{39}Ar$ and paleomagnetic data are effective criteria for determining ages, correlations, and stratigraphic relationships for ignimbrites in the Mogollon-Datil volcanic field. Several ignimbrites have been accurately correlated over distances of 40 to 200 km, providing reliable ties between previously established local and subregional stratigraphic sequences. The data are sufficient to allow established ignimbrite sequences to be assembled into a single, integrated time-stratigraphic framework for the entire volcanic field. Generalized subregional composite ignimbrite sequences that make up the time-stratigraphic framework are depicted in Figure 11, which includes a summary of measured $^{40}Ar/^{39}Ar$ ages and paleomagnetic polarity data. Figure 12 provides an even more generalized version of the Mogollon-Datil time-stratigraphic framework and compares this framework with pre-existing interpretations based primarily on lithologic correlations and conventional K-Ar data. Implications of the Mogollon-Datil time-stratigraphic framework are discussed below.

DISCUSSION

Application of $^{40}Ar/^{39}Ar$ Dating and Paleomagnetism in Silicic Volcanic Fields

Results of this and other studies (for example, Deino and Best, 1988; Deino, 1989) demonstrate that the combination of $^{40}Ar/^{39}Ar$ dating and paleomagnetic analysis is a powerful approach for addressing correlation problems in large silicic volcanic fields. Both techniques are well suited for study of regional ignimbrite outflow sheets. Silicic ignimbrites commonly contain sanidine or anorthoclase; these quenched, anhydrous K-feldspars offer inherently more precise $^{40}Ar/^{39}Ar$ age determinations than do hydrous mafic K-rich minerals or whole-rock material. In this study, $^{40}Ar/^{39}Ar$ data from sanidine separates of multiple samples from individual ignimbrites routinely attained within-unit precision (1σ) of ± 0.1% to 0.4%.

Precise paleomagnetic directions are also readily measured from silicic ignimbrites. In this study, 90% of sampled sites yielded tightly grouped remanence directions (within-site α_{95} = 1°–10°) interpreted as TRM acquired during rapid post-emplacement cooling. During this initial cooling, magnetic oxides in the ignimbrites experienced high-temperature deuteric oxidation, which produced magnetite-hematite assemblages resistant to chemical alteration and acquisition of secondary components other than easily removable lightning-induced isothermal magnetizations. Although accuracy of TRM data at some sites is affected by tectonic

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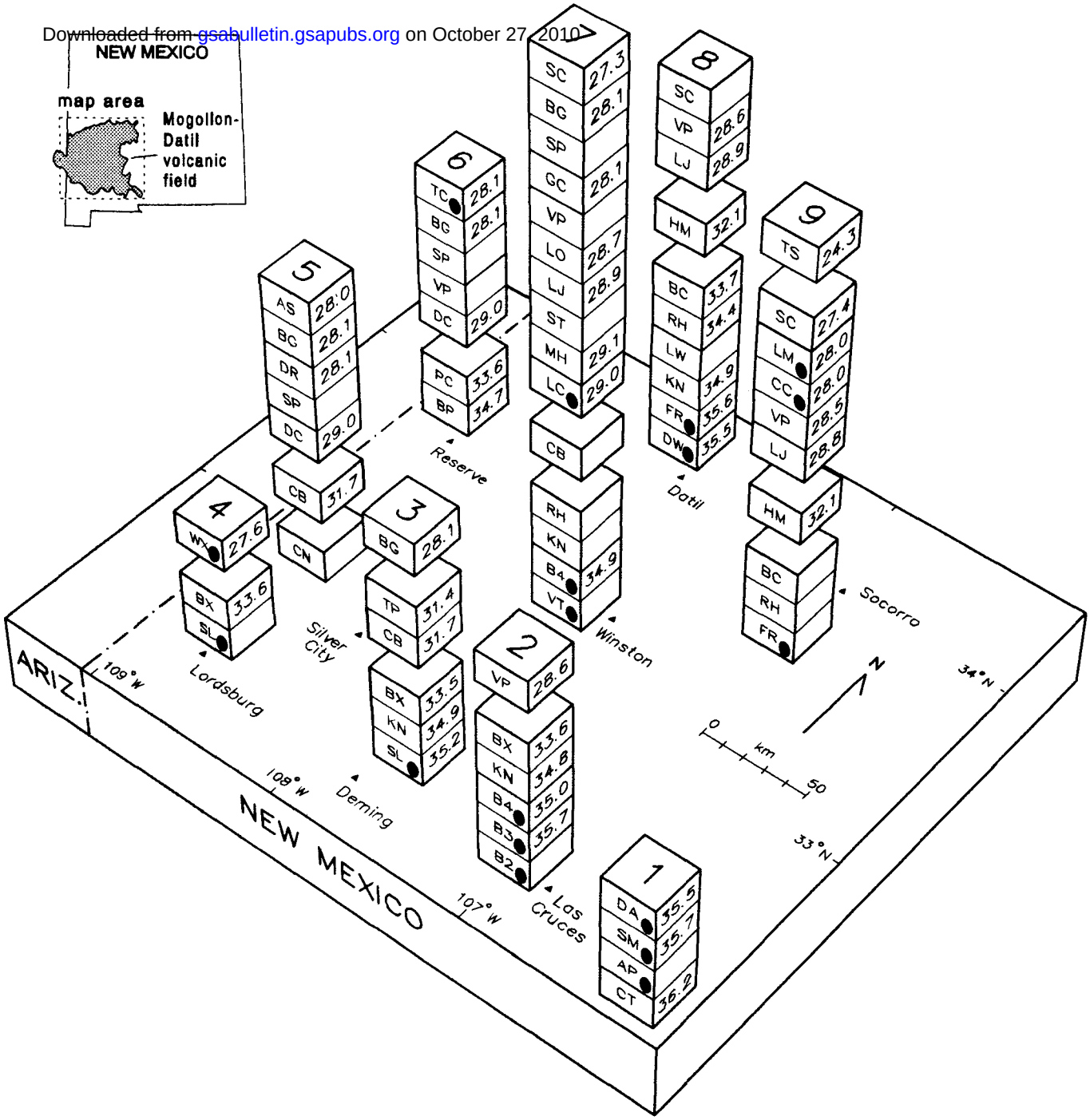
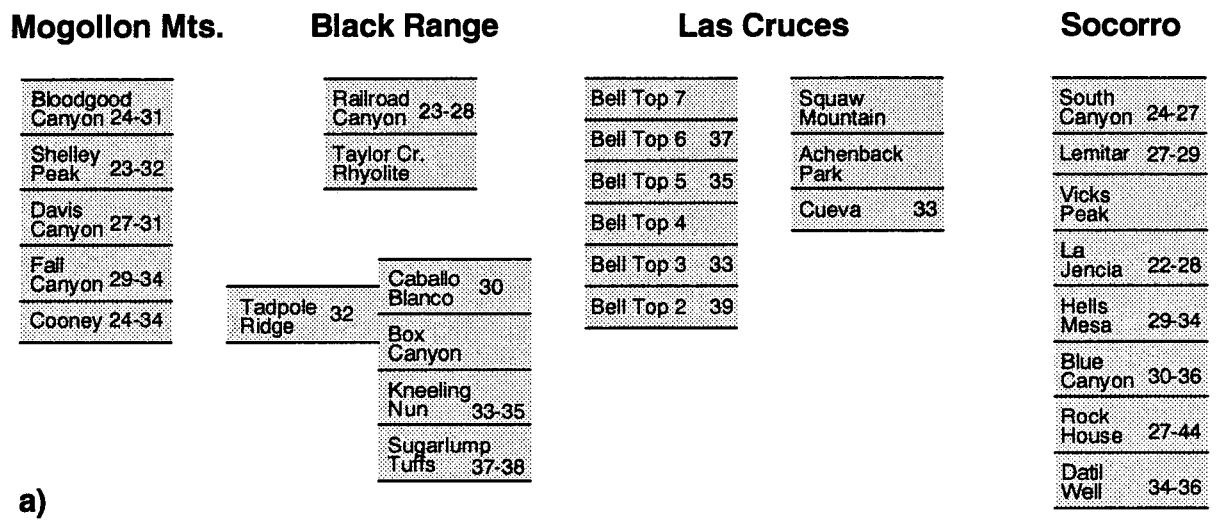


Figure 11. Three-dimensional view of time-stratigraphic framework for the Mogollan-Datil volcanic field. Columns represent generalized composite ignimbrite sequences for selected subregions of the Mogollon-Datil volcanic field. Eruptive hiatuses are shown as gaps in the columns. Column locations are approximate; most of the composite sequences are not present in their entirety at any one location. Blocks depict individual ignimbrites, showing the abbreviated unit name (corresponding to Table 1), paleomagnetic polarity (black dot = normal polarity; no symbol = reverse polarity), and $^{40}Ar/^{39}Ar$ age (some of these local data differ slightly from unit-mean ages presented in Table 1). Blocks lacking ages indicate that the unit was not dated in that area but was correlated using paleomagnetic data, stratigraphic position, and lithology. Abbreviations not shown in Table 1: BP = tuff of Bishop Peak, probable correlative of LW; VT = tuff of Victoria Tank, possible correlative of FR.

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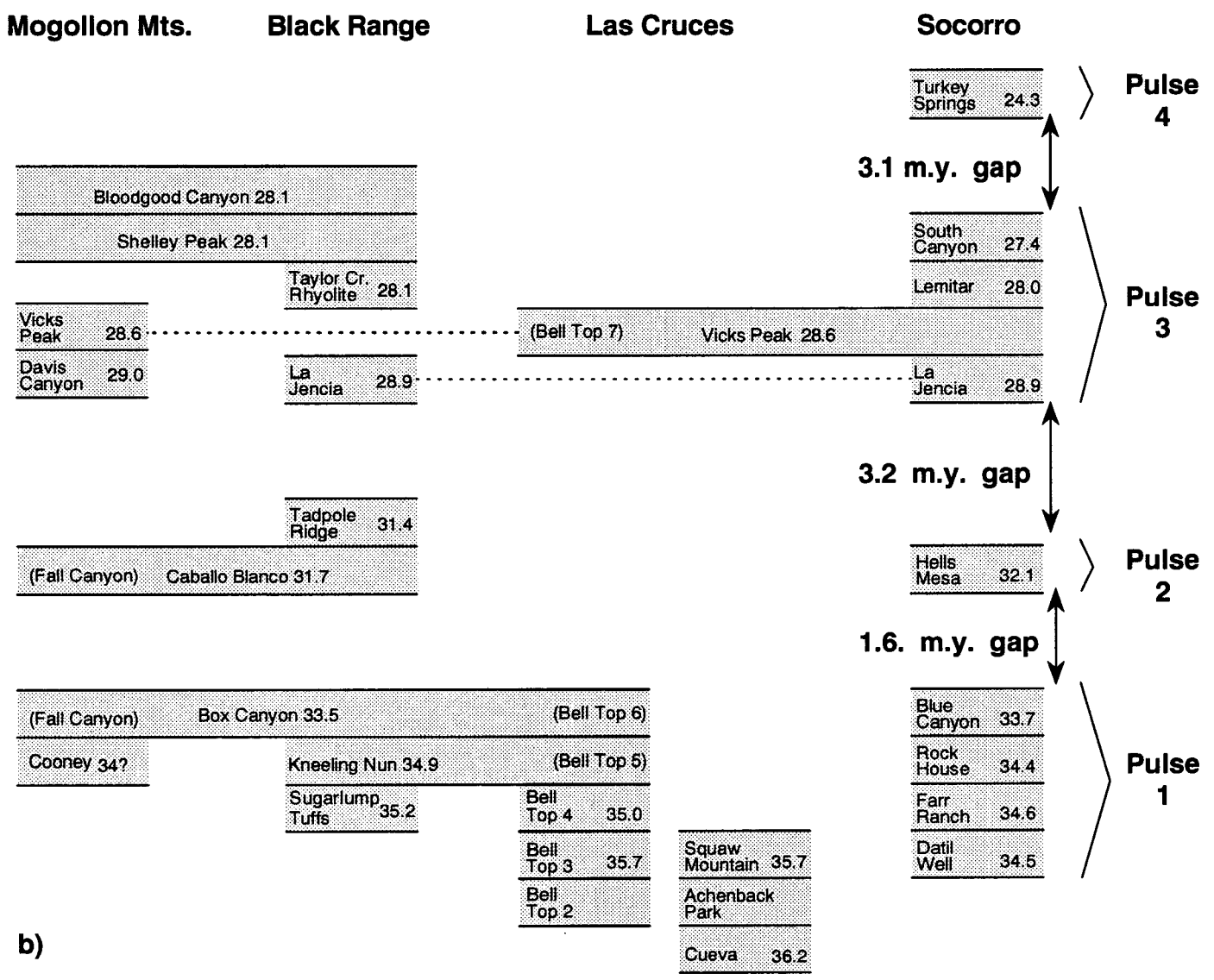


Figure 12. Evolution of the time-stratigraphic framework for major regional ignimbrites in the Mogollan-Datil volcanic field. (a) Subregional stratigraphic sequences determined prior to this study. Numbers indicate the range of published K/Ar data (Ma) rounded to the nearest million years. Sources: Ratté and others, 1984; Marvin and others, 1987; Clemons, 1976; Seager and McCurry, 1988; Elston, 1984; Osburn and Chapin, 1983a. (b) Revised generalized stratigraphic sequences for the same areas, showing $^{40}Ar/^{39}Ar$ ages, and correlations aided by $^{40}Ar/^{39}Ar$ and paleomagnetic data. Parentheses denote pre-existing names for correlated parts of some of the major ignimbrite outflow sheets. In addition to regional ignimbrites, the figure includes Taylor Creek Rhyolite lava.

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rotations and uncertainties in determination of paleohorizontal, the data are sufficent to reliably distinguish among many of the studied ignimbrites, including units of matching paleomagnetic polarity.

Numerous correlation problems in the Mogollon-Datil volcanic field have been successfully addressed using $^{40}Ar/^{39}Ar$ dating, paleomagnetic analysis, and existing lithologic and stratigraphic data. $^{40}Ar/^{39}Ar$ and paleomagnetic correlation criteria tend to be complementary. Where unaltered sanidine is available, $^{40}Ar/^{39}Ar$ data serve to resolve age differences between units in which unit-mean paleomagnetic directions are similar or site-mean directions have been disturbed by tectonic rotations (for example, Kneeling Nun, Box Canyon, Caballo Blanco, and Hells Mesa Tuffs; Figs. 5 and 7). Conversely, paleomagnetic data help to distinguish among units which lack sanidine or are too close in age to be resolved by the $^{40}Ar/^{39}Ar$ method (for example, Bloodgood Canyon Tuff, tuff of Triangle C Ranch, Lemitar Tuff, and South Canyon Tuff; Figs. 8 and 9).

Implications of the Time-Stratigraphic Framework for the Mogollon-Datil Volcanic Field

Because it is based on high-precision $^{40}Ar/^{39}Ar$ age data and reliable correlations, the Mogollon-Datil time-stratigraphic framework reveals features of volcanic activity that were not previously apparent. The most glaring feature is the strongly episodic nature of ignimbrite volcanism, clearly apparent in the $^{40}Ar/^{39}Ar$ data (Figs. 11 and 12b), but never recognized using conventional K-Ar data (Fig. 12a). A second example is the interpretation of ignimbrite members of the Bell Top Formation, exposed in a basin north and west of Las Cruces. The Bell Top Formation tuffs (Fig. 5) were previously interpreted to be locally derived ignimbrites (Clemons, 1976). This interpretation led to the conclusion that the basin in which these ignimbrites accumulated was a volcanotectonic depression produced by repeated episodes of local magma withdrawal (Seager, 1973). $^{40}Ar/^{39}Ar$ and paleomagnetic data demonstrate that all but the lowest two Bell Top Formation tuffs are actually distal facies of ignimbrites erupted elsewhere, clearly showing that magma dynamics were not responsible for basin subsidence.

The time-stratigraphic framework for the Mogollon-Datil volcanic field has numerous potential applications. Precise ages for the individual ignimbrites should aid geochemical modeling of magmatic evolution. As an example, $^{40}Ar/^{39}Ar$ data can be directly applied to calculations of initial $^{87}Sr/^{86}Sr$ ratios. Alternatively, studies of magma-chamber evolution are enabled by reliable identification of units erupted close enough in space and time to be potential products of a common magma chamber. Examples of probable comagmatic pairs of ignimbrites, previously unrecognized as such, include Bell Top Formation tuff 4 and Kneeling Nun Tuff, Caronita Canyon Tuff and Lemitar Tuff, and Bloodgood Canyon Tuff and tuff of Triangle C Ranch.

Another potentially useful aspect of the time-stratigraphic framework is the capability to reliably trace individual ignimbrites over their entire exposed extent. This opens the door to a host of other studies ranging from detailed examination of radial and sectoral geochemical variations within individual outflow sheets to studies of paleotopography or physical volcanology. In addition, stratigraphic bracketing by precisely dated ignimbrites provides firm age control for interlayered sedimentary or volcanic rocks which cannot be precisely dated themselves. In this way, the time-stratigraphic framework can be used to enhance radiometric calibration of biostratigraphy or to provide a test bed for improving methods of dating mafic rocks. As an example of useful stratigraphic bracketing, the position of the Poverty Creek basaltic andesite lavas beneath the 29.0 Ma tuff of Little Mineral Creek (Table 1) clearly demonstrates that these lavas are older than their published K-Ar ages (Elston and others, 1973).

Finally, the Mogollon-Datil time-stratigraphic framework and paleomagnetic data set have important applications for calibration of the worldwide geomagnetic polarity time scale; this aspect is discussed elsewhere (McIntosh, 1988).

Mogollon-Datil Ignimbrite Activity

The details of ignimbrite activity in the Mogollon-Datil volcanic field have been presented in previous sections; the following paragraphs highlight a few aspects of this activity.

The most striking feature of this time-stratigraphic framework is the strong episodicity of ignimbrite activity, which was confined to four brief (<2.6 Ma) eruptive pulses separated by 1- to 3-m.y. gaps during which no cauldron-forming eruptions occurred (Figs. 11, 12, and 13). The timing, volumes, and mean effusion rates of the eruptive pulses are summarized in Figure 13. The abrupt turning on and off of ignimbrite pulses was synchronized between the northern and southern cauldron clusters in the field, particularly during pulse 3 (Fig. 12 and Table 1). Furthermore, the initiation of pulse 3 near 29.0 Ma corresponded closely to initiation of the main pulse of ignimbrite activity in the San Juan volcanic field (Lipman and others, 1978). The regionally synchronous nature of initiation of episodes of ignimbrite volcanism suggests that a major controlling factor may be regional crustal stress, probably related to plate-tectonic interactions along the western margin of the American plate. Further discussion of this topic is beyond the scope of this paper and will be pursued elsewhere.

A second striking feature of Mogollon-Datil volcanic activity is the northwest to west migration of eruptive centers (Fig. 2). Activity in the southern complex began at 36.2 Ma near Las Cruces, New Mexico, and subsequently migrated 220 km northwest, culminating in the 28.0 Ma Bursum cauldron. Activity in the northern complex, located west of Socorro, New Mexico, underwent a less defined and more modest, 40-km westward migration over its 32.0 to 24.3 Ma life span. Interpretation of this pattern of cauldron migration is presently uncertain, although it fits crudely into the general westward sweep of mid-Tertiary volcanism long recognized in the southwestern United States (for example, Cross and Pilger, 1982).

Finally, an intriguing pattern of rhyolitic dome activity is revealed by the Mogollon-Datil time-stratigraphic framework. Although much of the effusive rhyolite activity in the Mogollon-Datil field was spatially associated with cauldrons, large volumes of rhyolite lava were erupted during ignimbrite pulse 3 in a cauldron-free area in the northern Black Range, directly between the Socorro and Mogollon cauldron clusters. Dating of lavas and associated ignimbrites, together with stratigraphic bracketing by dated regional ignimbrites, shows that these lavas were erupted in two

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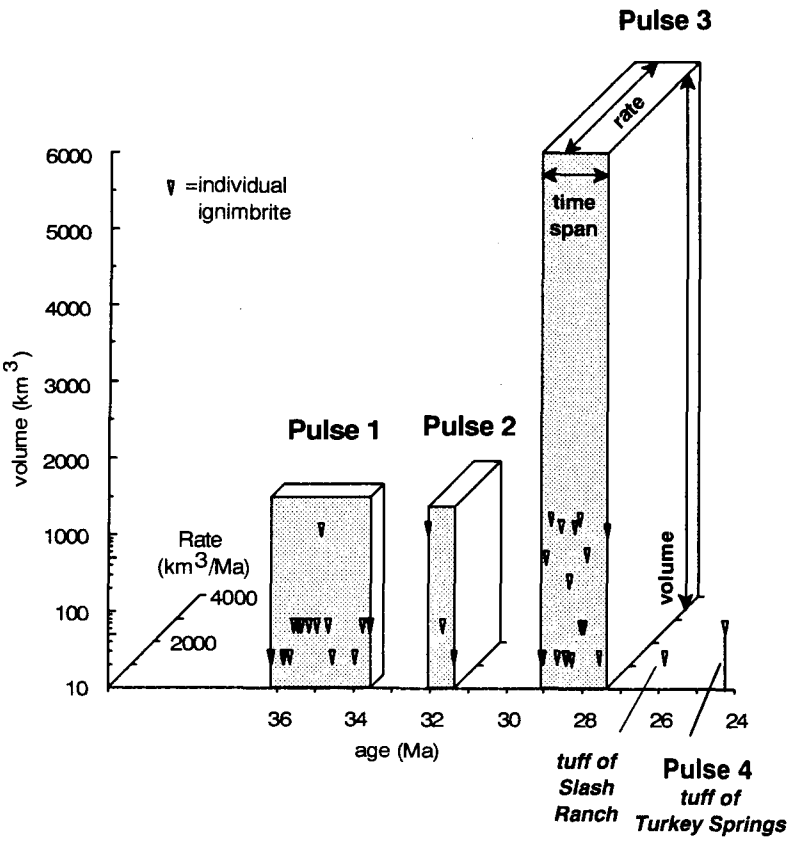


Figure 13. Timing, volume, and mean effusion rate for ignimbrite eruptive pulses in the Mogollon-Datil volcanic field, showing strongly episodic nature of volcanism. Triangles represent individual dated ignimbrites, with estimated volumes from Table 1. Units with no available volume estimate are conservatively shown as 50 km^3 for regional units (maximum known lateral dimension >40 km) and 20 km^3 for more local units.

brief, discrete pulses, the first ranging from 29.0 to 28.7 Ma, and the second, the well-known Taylor Creek Rhyolites (Duffield and others, 1987; Dalrymple and Duffield, 1988), ranging over an even shorter interval near 28.1 Ma. Eruption of these rhyolites immediately preceded synchronized ignimbrite pulses in the adjacent cauldron clusters to the northeast and southwest (Fig. 2). Perhaps the rhyolite lavas were produced by a large underlying magma body responding to the same crustal stress responsible for the episodes of ignimbrite volcanism, but the condition of the crust in the northern Back Range was such that rising magmas were able to degas passively prior to eruption.

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